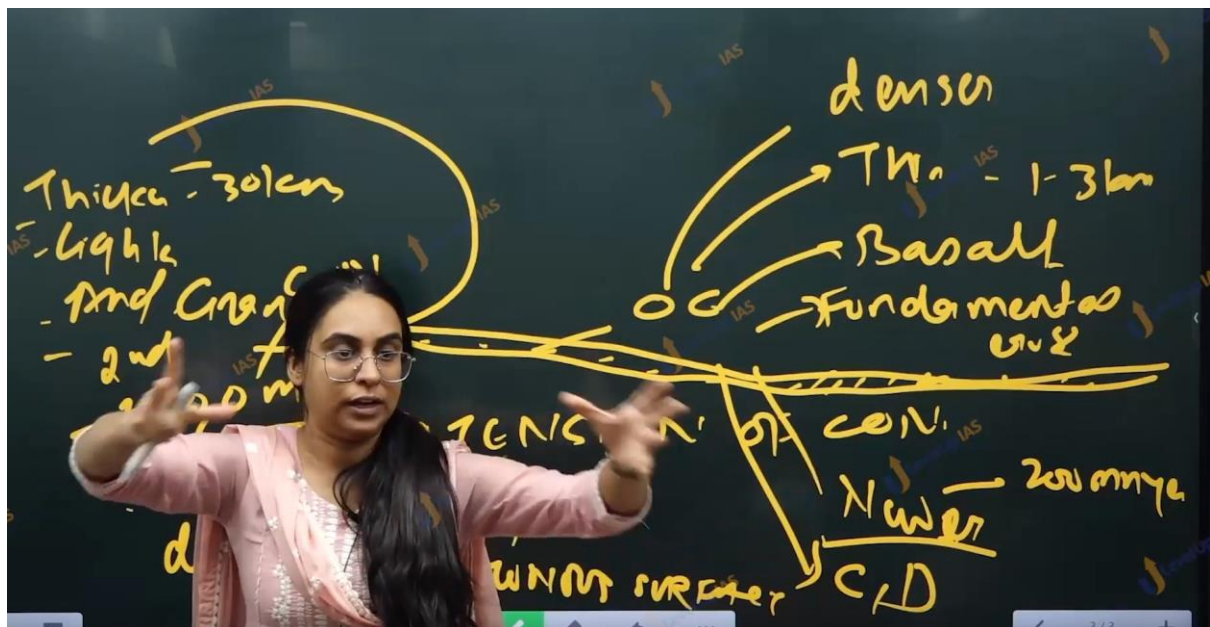
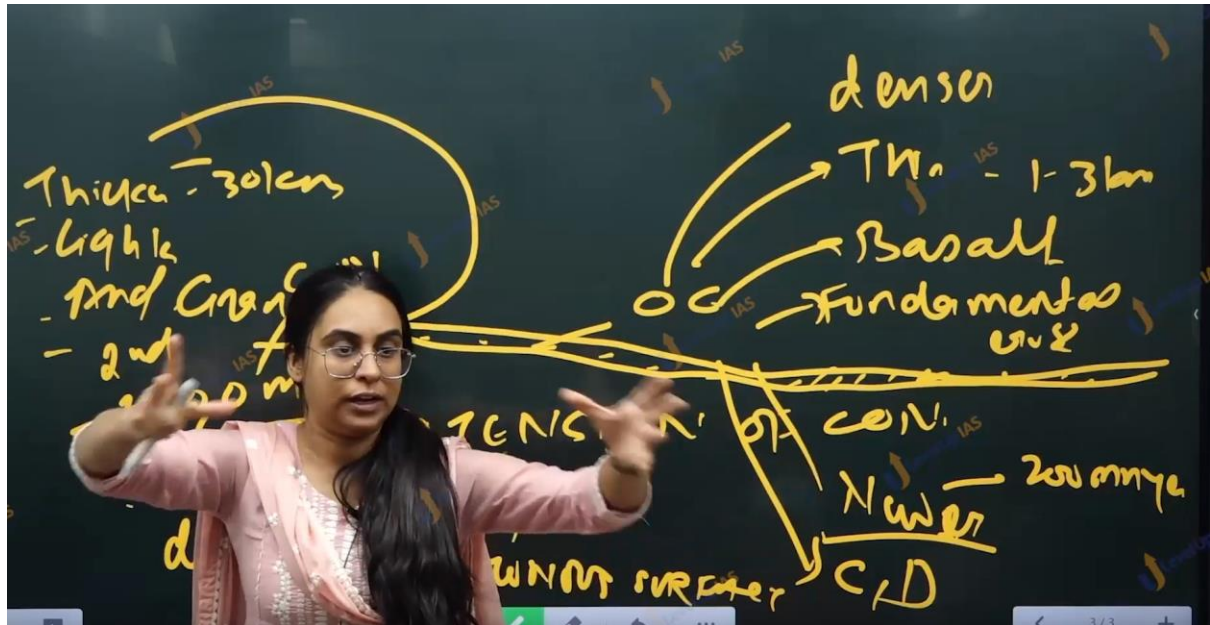


(24) Ocean Temperature & Salinity





LevelUp IAS

Chemical layering – 3 layers: Crust

CRUST

1. is the **outermost layer** and also the **lightest** among all.
2. On an average thickness in 17-20 km.

Together the continents and oceans are called **Relief Feature of 1st Order**.

On the Crust, Landforms can be classified into different Order:

1. **Relief Feature of 1st Order:** Continent and Oceans
2. **Relief Feature of 2nd Order:** Structural features developed over a continent or a ocean: Mountains, plateaus, lakes. These forms owe their genesis mainly to **endogenetic forces**. These are called **construction landforms**
3. **Relief Feature of 3rd Order:** Micro level landforms developed on the second order relief features by exogenic process. landforms may be **erosional** (glacial valley, River Valley) or **depositional** (drumlin, esker)

FIRST ORDER OF RELIEF

SECOND ORDER OF RELIEF

are those found on the continents and ocean basins.

THIRD ORDER OF RELIEF

Land and Water Features

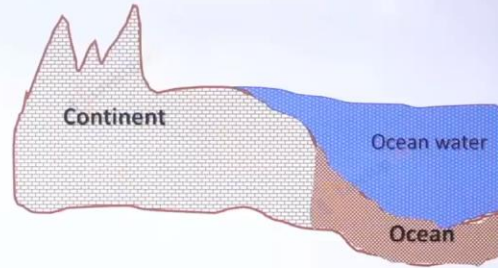
Volcano

Mountain

Chemical layering – 3 layers: Crust

CRUST

1. is the **outermost layer** and also the **lightest** among all.
2. On an average thickness in 17-20 km.
3. It can be divided into Continental Crust and Oceanic crust.

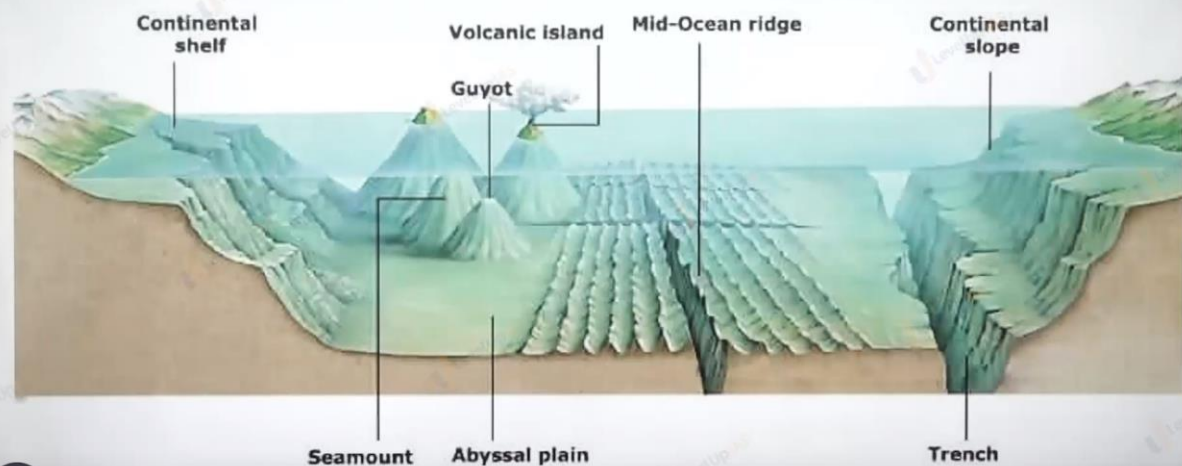


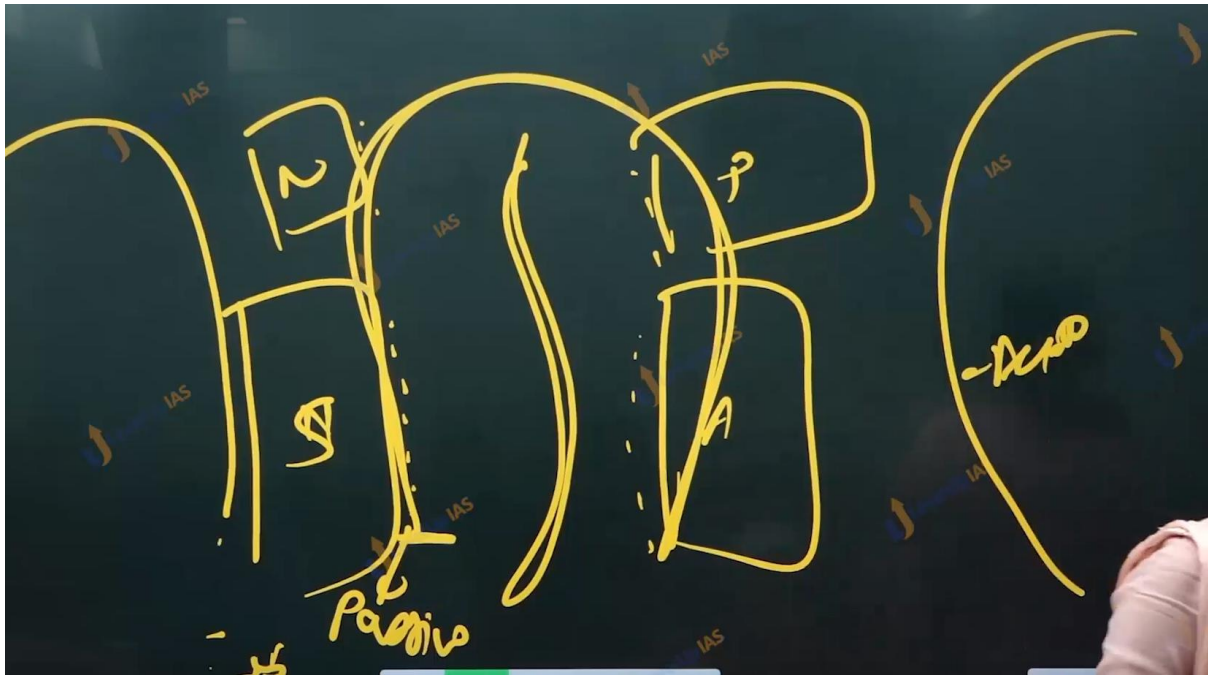
Continental Crust and Oceanic Crust

Continental Crust	Oceanic Crust
Thickness: 35-60 kilometre	Thickness: 1-3 kilometre
Density: 2-2.77 gm/ cm ³ so light	Density: 3-3.5 so heavier as compared to continent
Rocks: Granitic, Andesite, Diorite	Rocks: Basalt and <u>Gabro</u>
They are acidic	They are basic
Rocks are lighter in colour	Rocks are darker in colour

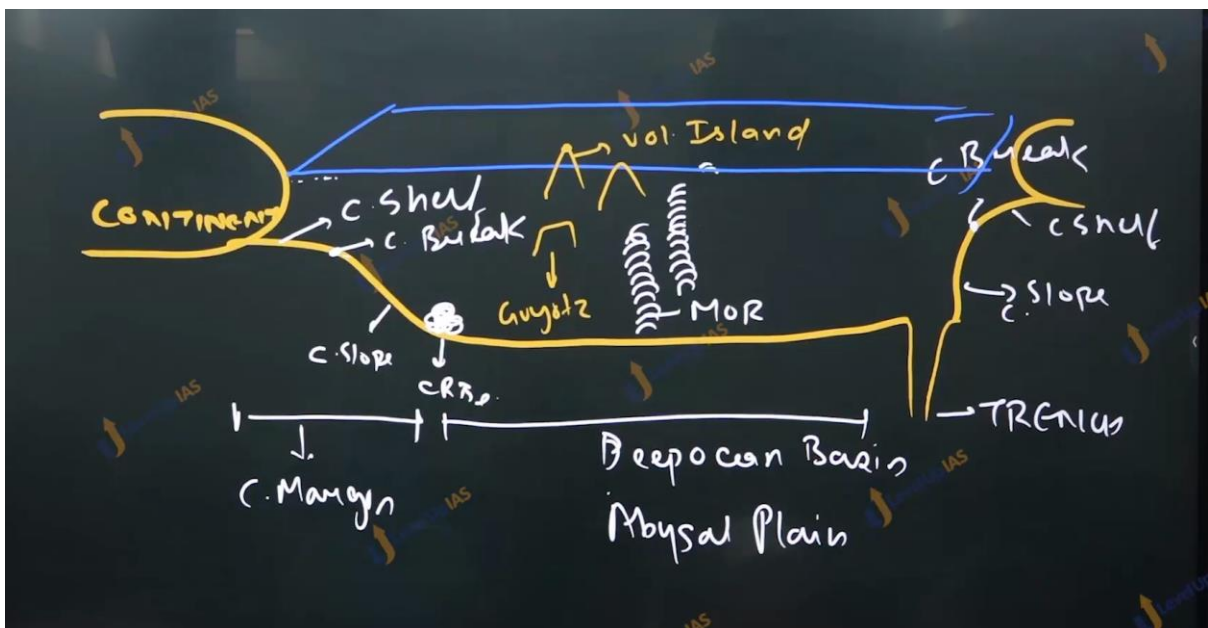
Ocean Bottom Topography

Ocean is not submerged extension of continent





Ocean bottom Relief:





1. The ocean floor are rugged with the largest mount range, deepest trench and largest plane. On the ocean floor the feature has been by the tectonic process and volcanic process depositional process.
2. Deeper point of trench is ocean deep
- 3.

Continental shelf:

1. It is a gently sloping extension of a continental plane.
2. The continental shelf of all the ocean together covers 7.5 % of the total area of ocean.
3. The depth of the continental shelf is 30 – 600 m. The average depth is 200m.
4. The gradient of the continental shelf is 1 degree or even less.
5. The end of continental shelf is called continental break. And the continental shelf is followed by a deep and steep slope.
6. The Continental shelf is covered by sediment brought by river and glacier.
7. Massive sedimentary deposits receive over the long time trap the marine organisms which eventually turns the source of fossil fuel.

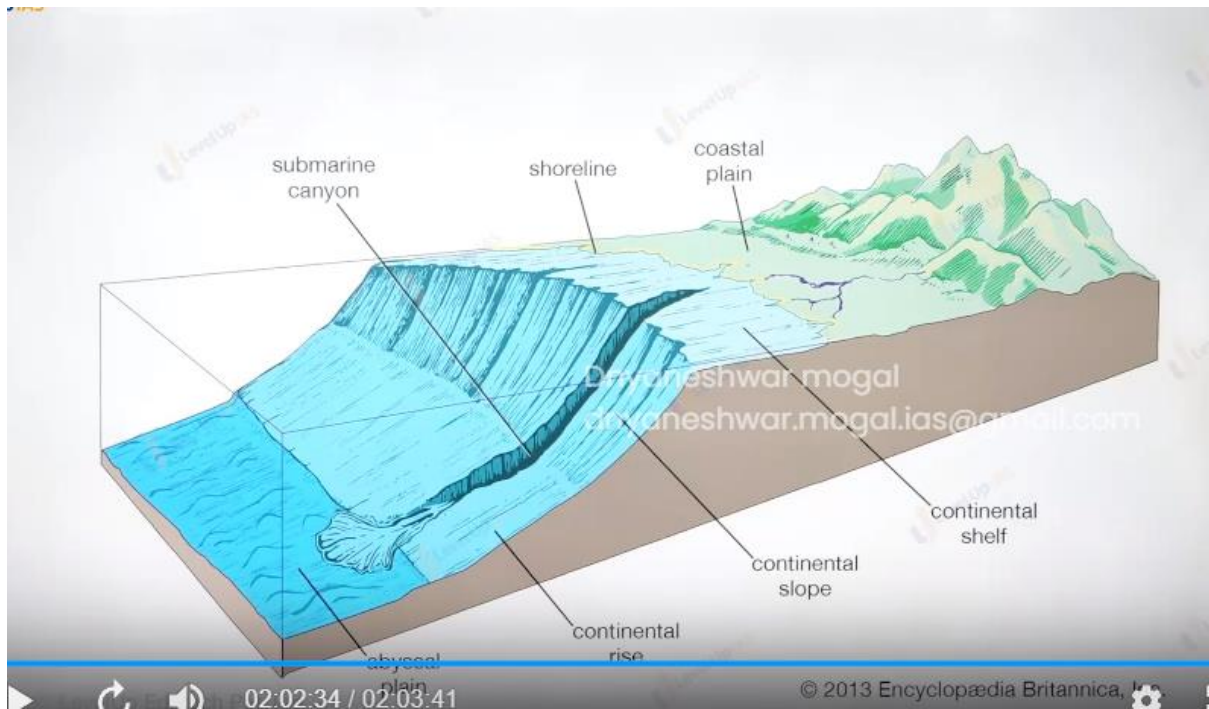
8. The Continental shelf is mainly formed due to submergence of part of continent also because of sedimentary brought by river.
 9. The average width is 80 km however continental shelf is almost absent or very narrow along the active margin (where plate interaction is taking place) it extensive when it on passive margin.
-

Continental slope.

-
1. Continental slope is slope between outer edge of continental shelf and deep ocean floor.
 2. It is cut by submarine canyons in many locations.
 3. The gradient of slope can varies 2 – 5 degree as much 30 -60 degree.
 4. The depth of continental slope 200 to 3500,3600 m. average depth of ocean floor is 4100 m.
 5. Continental slope because of steepness have very little deposit along the base of continental slope there are deposits of sediments called as continental rise this is seen along the passive margin.
 6. In the active margin rather the continental rise there is trench.
-

Submarine canyon:

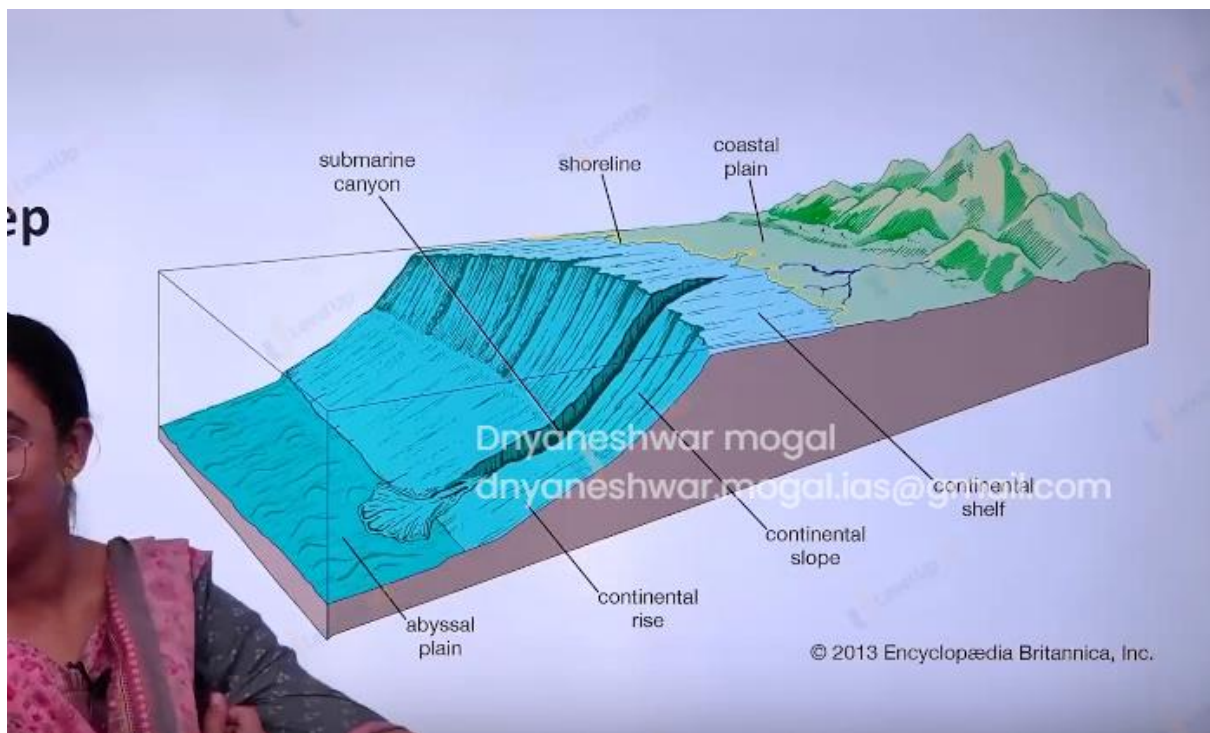
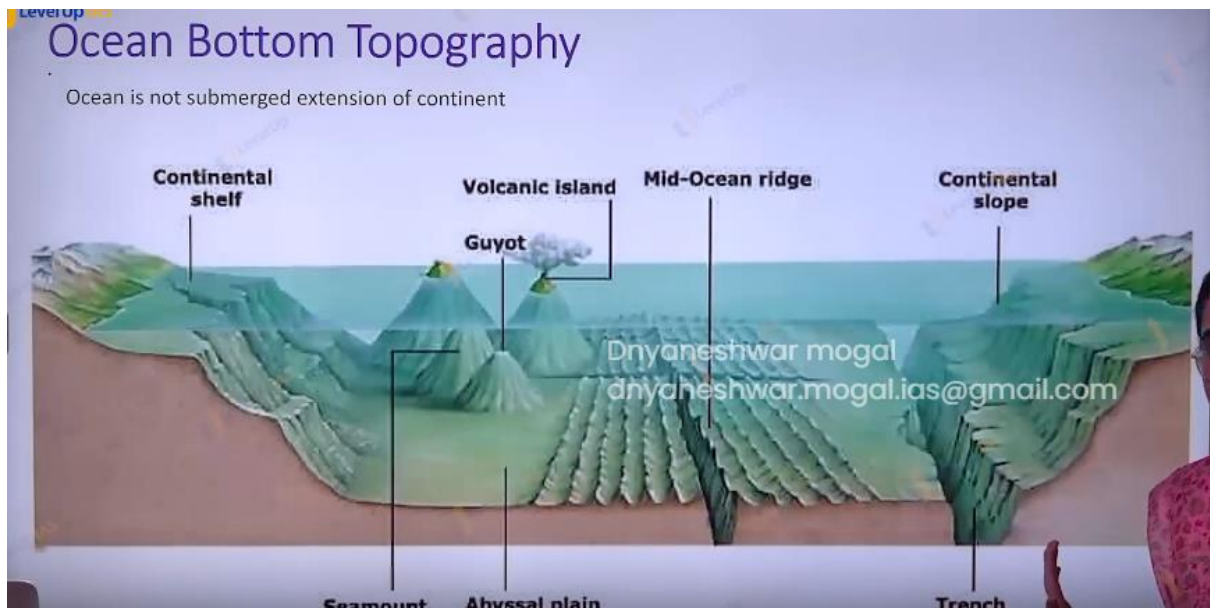
-
1. Submarine canyon are deep valleys inside the ocean water. They cut across the continental shelf and continental slope.
 2. They are founded by turbidity current (sediments falling off in the ocean water.) falling of from the continental shelf.
 - 3.
-

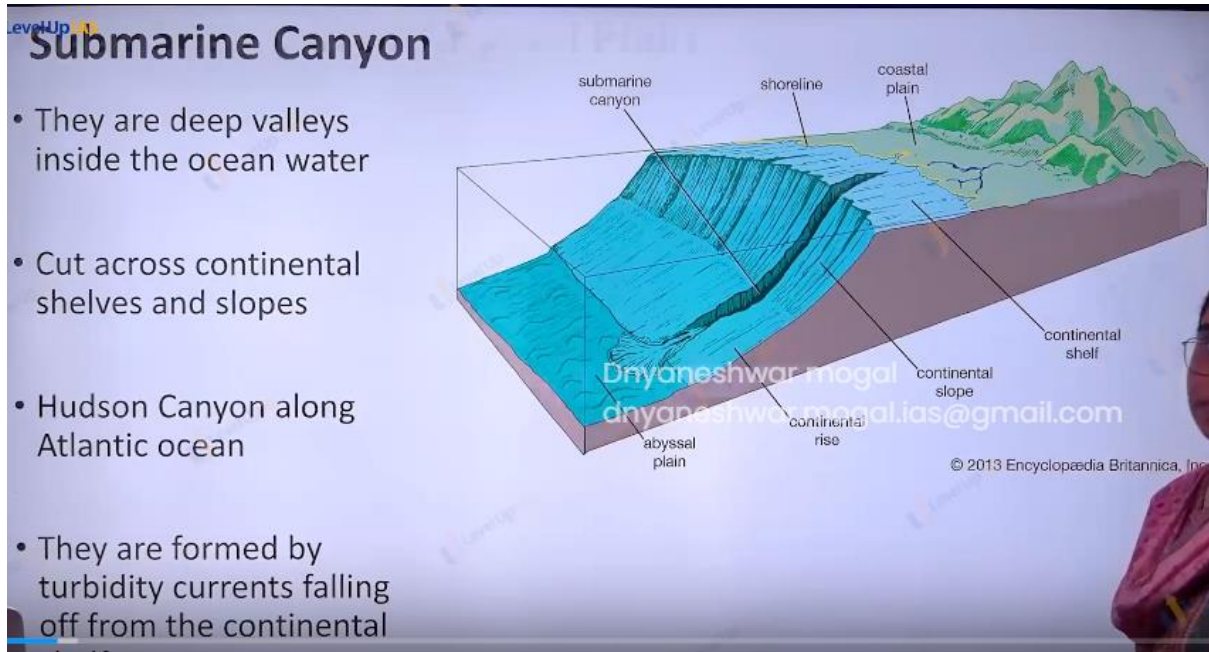


Abyssal plain/ deep ocean basin:

-
1. They are flattest surface in the ocean floor they are formed because of sediments eroded from the land and the deposit of ocean deposits.
 2. Depth 3000 – 4000m . deep ocean basin have extensive mountain chain (mid-oceanic ridge MOR), Oceanic islands, guyot,
-

(images of Ocean bottom topography)





Mid-Oceanic ridge

1. MOR is found on the divergent plate boundary. It is composed of chain of mountains.
2. The mountain ranges can have the peak up to 2500 km but sometime the peak can reach up to ocean surface. Ex island,
- 3.

Mid Oceanic Ridge

- A mid-ocean ridge is a seafloor mountain system formed by plate tectonics.
- This feature is where seafloor spreading takes place along a **divergent plate boundary**.
- Mid-oceanic ridge is composed of chains of mountains.
- Mountain ranges can have peaks as high as 2,500 m and some even reach above the ocean's surface.
- Iceland, a part of the Mid Atlantic Ridge, is an example.



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(images of Mid Oceanic Ridge)

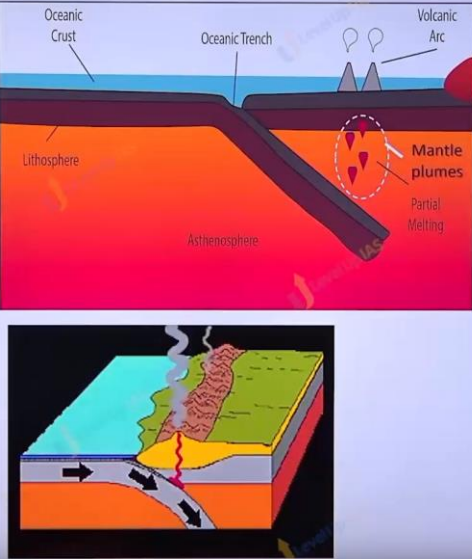
Oceanic Deep or Trenches

- The trenches are very common in the Pacific Ocean and form an almost continuous ring along the western and eastern margins of the Pacific.
- Ocean deep is deepest point of

Trench in the Mariana Sea is the deepest trench more than 11km

associated with active volcanism

have been 32 are the



Trenches and the ocean deeps.

1. Trenches are relatively steep and narrow gaps on the ocean floor.

2. Trenches can be 3-5 km deeper than the surrounding ocean floor.
(they are some of the deepest part of ocean)
 3. Trenches are formed due to ocean-ocean convergent and subduction or Ocean-continental convergent and subduction.
 4. Ex: Mariana, Philippine, common along the Pacific Ocean.
 5. Trenches are very common in the Pacific Ocean ex.

 - a. Aleutian trench
 - b. Kuril trench
 - c. Japan trench
 - d. Izu Bonin trench
 - e. Mariana's trench
 - f. Bougainville trench
 - g. Tonga trench

 6. Mariana trench is deepest trench with depth more than 11 km.
 7. Ocean deep is deepest point of trench ex. Challenger deep.
-

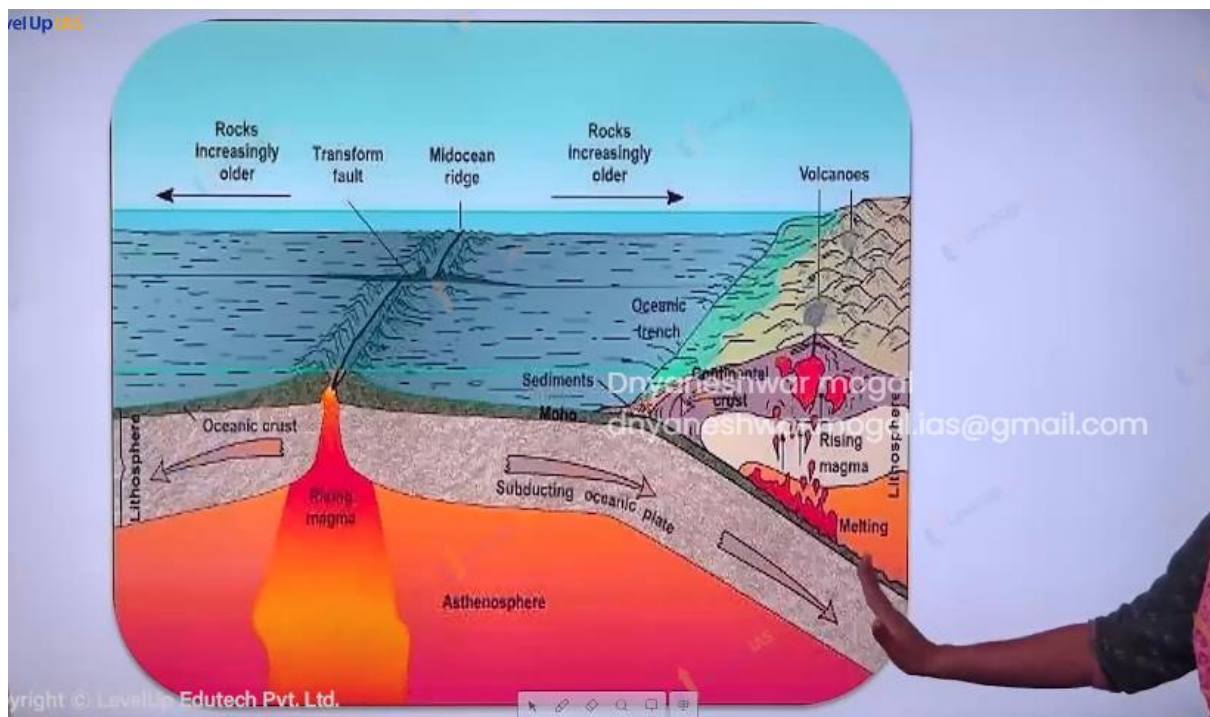


(ring of fire images)

(image at 25:30)

Seamount, Volcanic Island and Guyot:

1. They are generally formed due to hot spot activity. Volcanic island can reach up to surface, where sea mount does not reach the surface.
2. Sea mount has a pointed summit (top).
3. Sea mount is formed because of volcanic activity and can be as tall as 3000 m. ex emperor sea mount.
- 4.

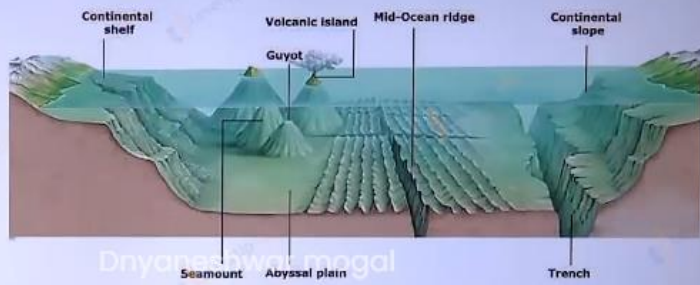


(image at 26:52)

Seamount, Volcanic Island and Guyot

Sea Mount

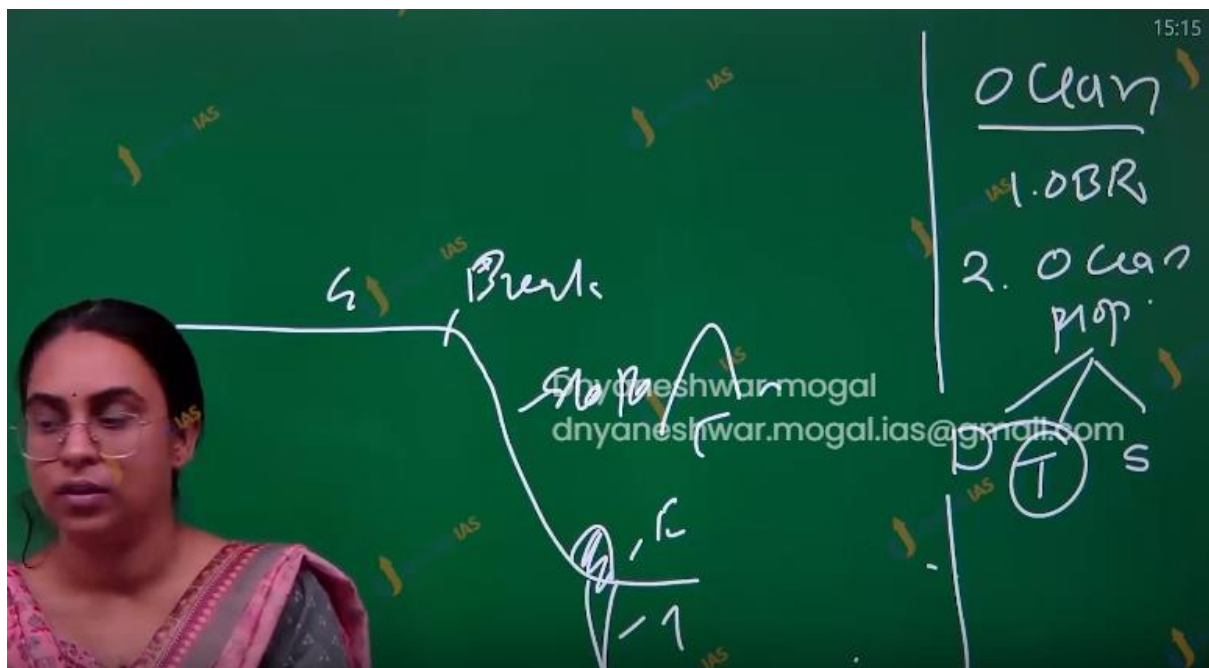
- A seamount is an **underwater mountain with pointed summit**
- Seamount **does not reach the surface of the ocean.**
- Seamount is formed by **volcanic activity.**
- These can be 3,000-4,500 m tall.
- **Emperor seamount**, an extension of the Hawaiian Islands in the Pacific Ocean, is a good example



Guyots:

(images at 30:39)

-
1. Guyots are table mounts. it is flat top sea mount.
 2. It shows gradual subsidence due to erosion and therefore it flat top.
 - 3.
-



(images at the 34:12)

(images at 35:40)

Ocean Temperature

- Average Ocean Temperature= 3.5 degree Celsius

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Business News | News | Science | Ocean's average temperature is 3.5 degree Celsius

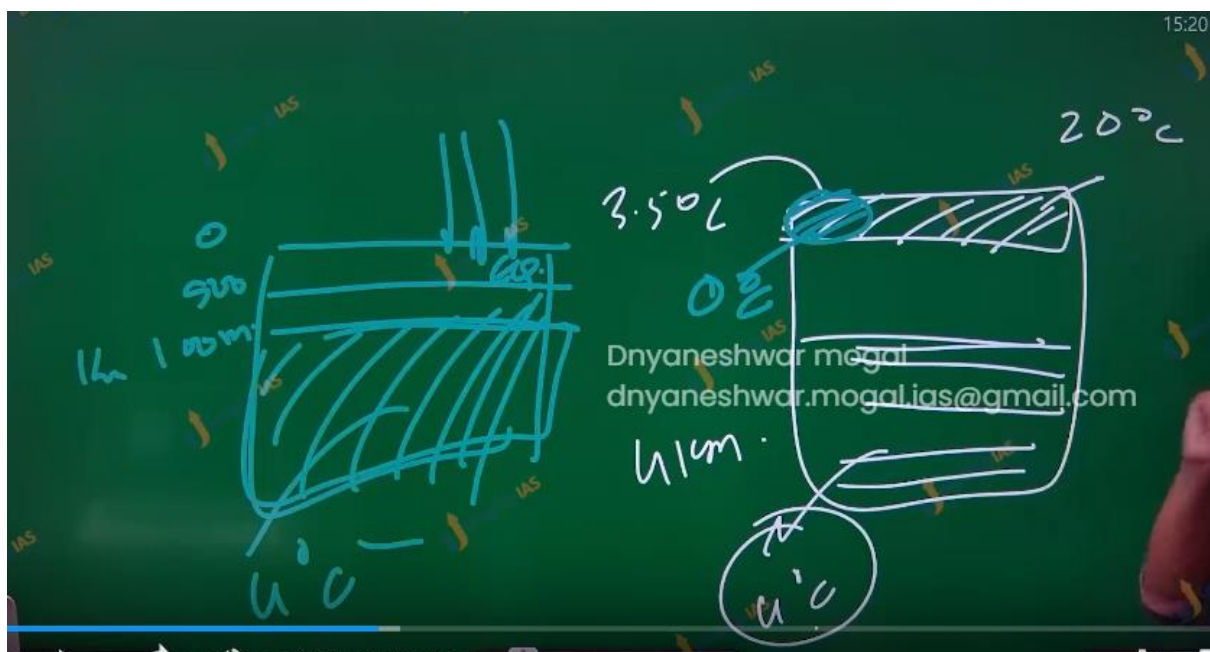
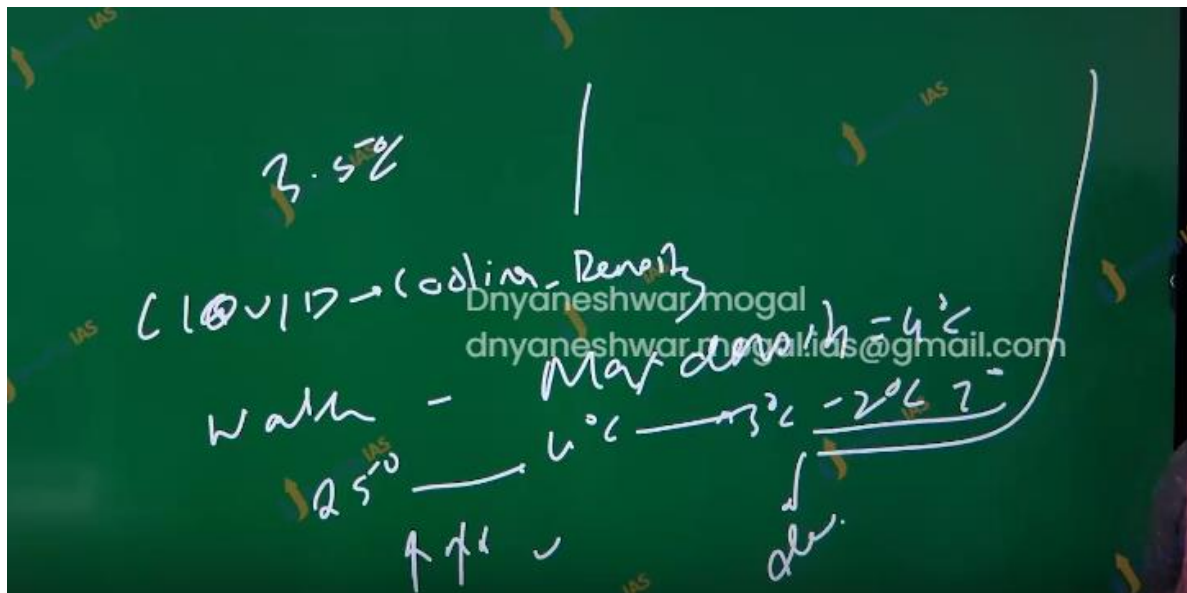
Ocean's average temperature is 3.5 degree Celsius

Last Updated: Jan 04, 2018, 12:52 PM IST

How does the temperature of ocean water vary?

The temperature of ocean water varies by location – both in terms of latitude and depth, variations in solar radiation

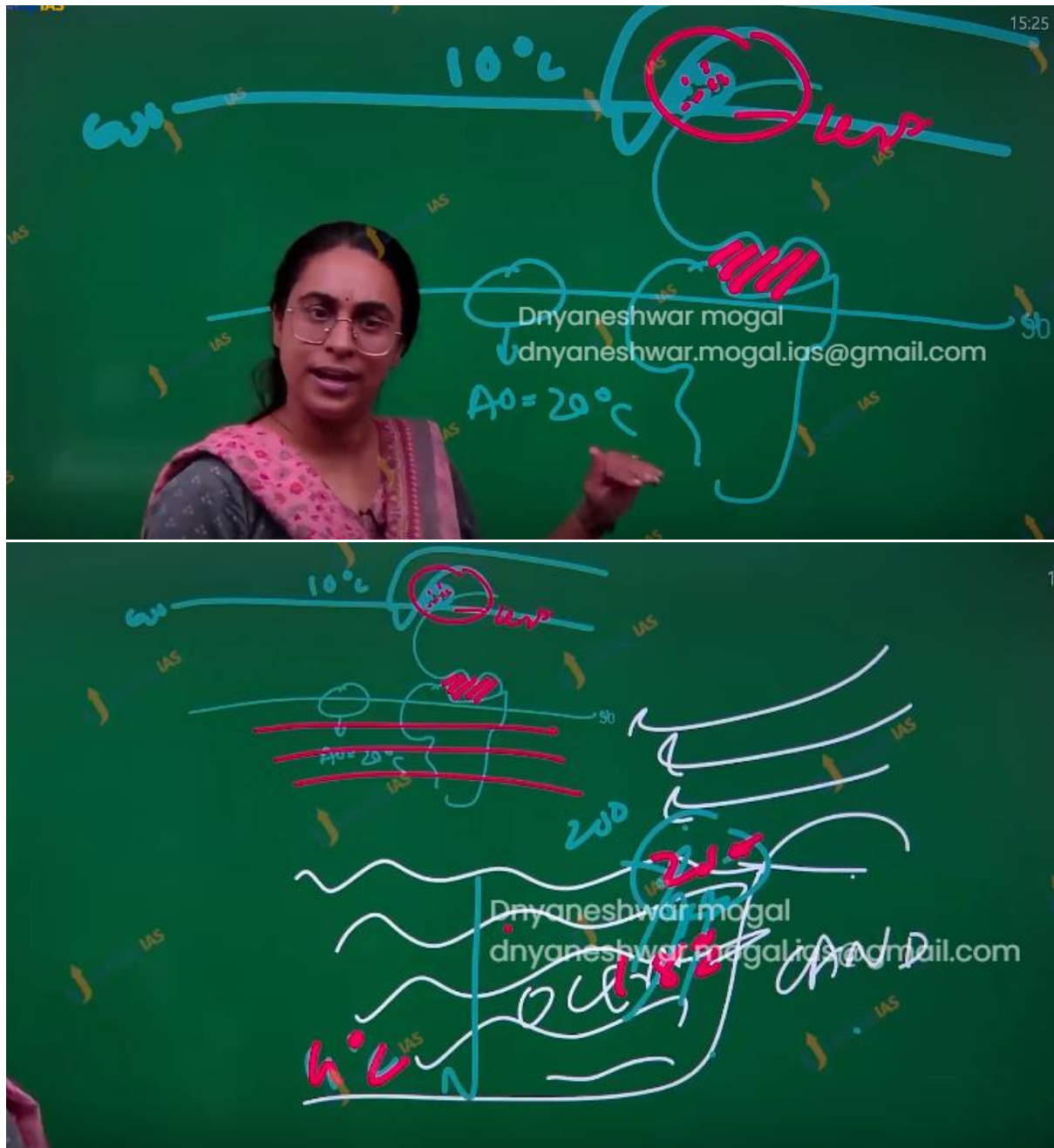
35:49



Ocean Temperature:

1. The average ocean temperature is about 3.5 degrees Celsius, however the temperature vary by the location both w.r.t latitude and depth. Due to variation in the solar radiations.

Factor affecting the Ocean temperature:



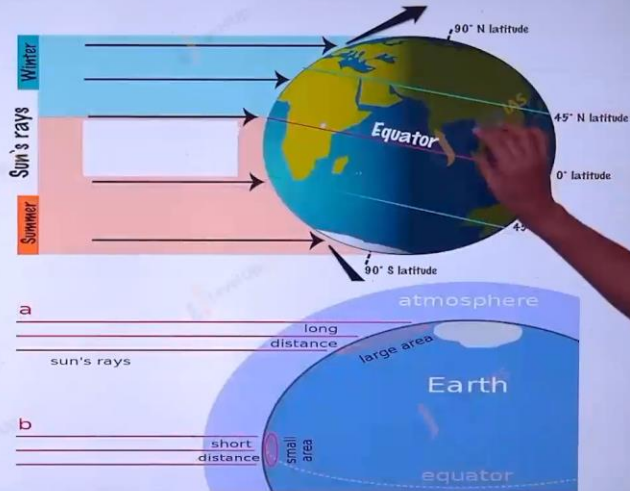
-
1. Latitude or insolation: Ocean gets heated by the solar energy therefore temperature of surface water decreases from tropics to poles because the amount of insolation decreases the pole wards.
-

Factors affecting temperature of Oceans

Ocean waters get heated up by the solar energy

Factors affecting Ocean Temperature

- (i) **Latitude** : Temperature of surface water decreases from the tropics towards the poles because the amount of insolation decreases poleward



Factors affecting temperature of Oceans

- (ii) **Unequal distribution of land and water:**

The oceans in the northern hemisphere receive more heat due to their contact with larger extent of land than the oceans in the southern hemisphere.

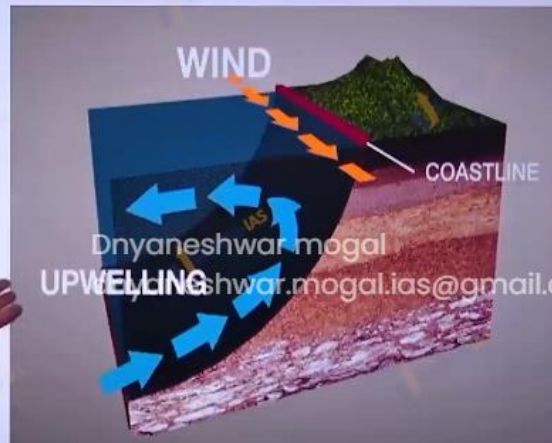


Irregular isotherms in North

Factors affecting temperature of Oceans

(iii) Prevailing winds:

Offshore trade winds drive warm surface water away from the coast resulting in the upwelling of cold water from below. It results in the longitudinal variation in temperature.



1. Latitude or insulation:
2. Unequal distribution of land and water: the Ocean in the northern hemisphere has more temperature than the ocean that southern hemisphere.
3. Prevailing winds: offshore trade winds dry the warm surface water away from the coast resulting in upwelling of cold water from below. Contrary to this onshore the pull up the warm water near the coast thus raising the ocean temperature.
4. Ocean currents :
 - a. Warm ocean currents raised the temperature ex gulf stream raised the temperature of western side of Europe.
 - b. Cold ocean currents reduces the temperature of the place. Ex Labrador ocean currents.
5. The condition in the enclosed sea.
 - a. Enclosed sea in the lower latitudes records relatively higher temperatures than the open ocean. Whereas enclosed sea in the higher latitudes records lower temperatures than the open sea.

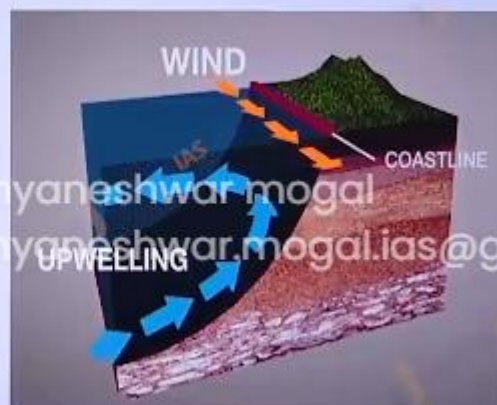


Factors affecting temperature of Oceans

(iii) Prevailing winds:

Offshore trade winds drive warm surface water away from the coast resulting in the upwelling of cold water from below. It results into the longitudinal variation in the temperature.

Contrary to this, the onshore winds pile up warm water near the coast and this raises the temperature.



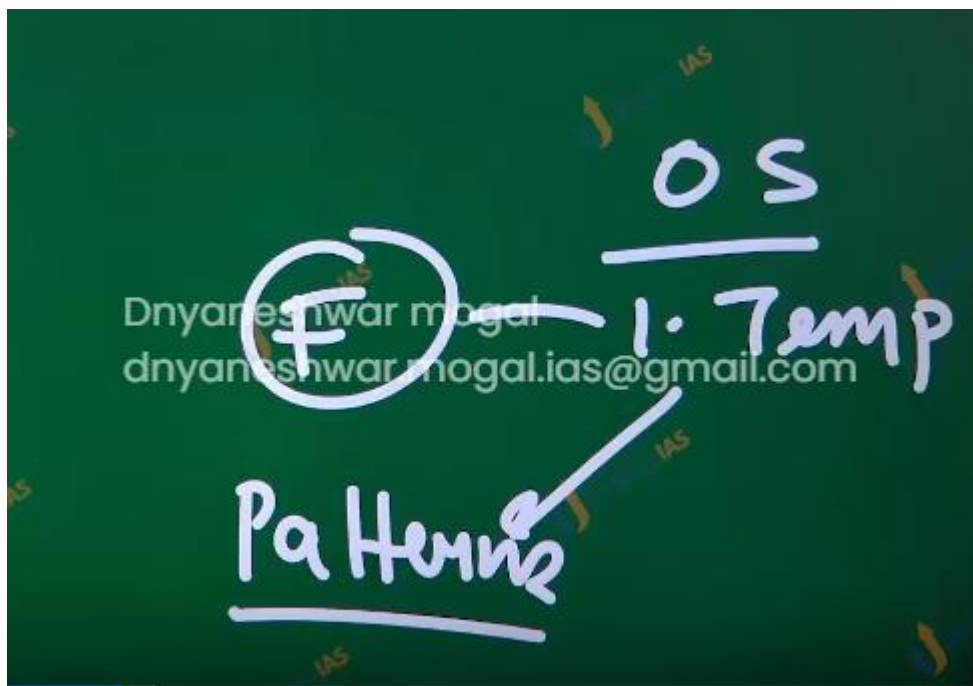
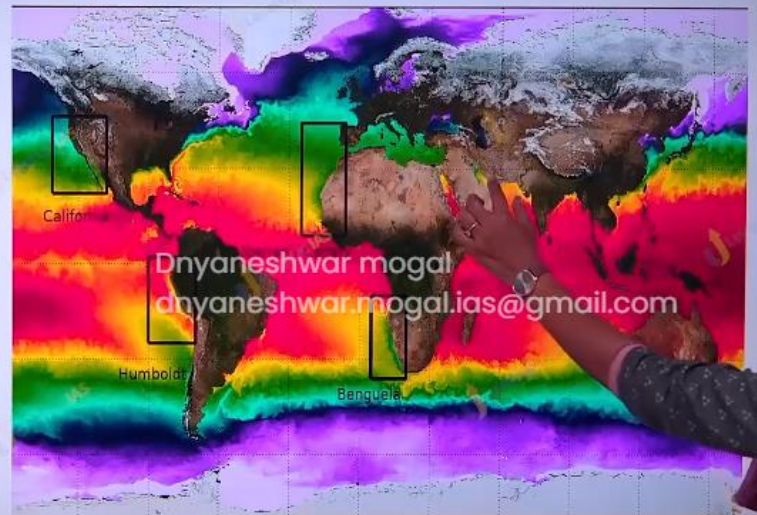
Factors affecting temperature of Oceans

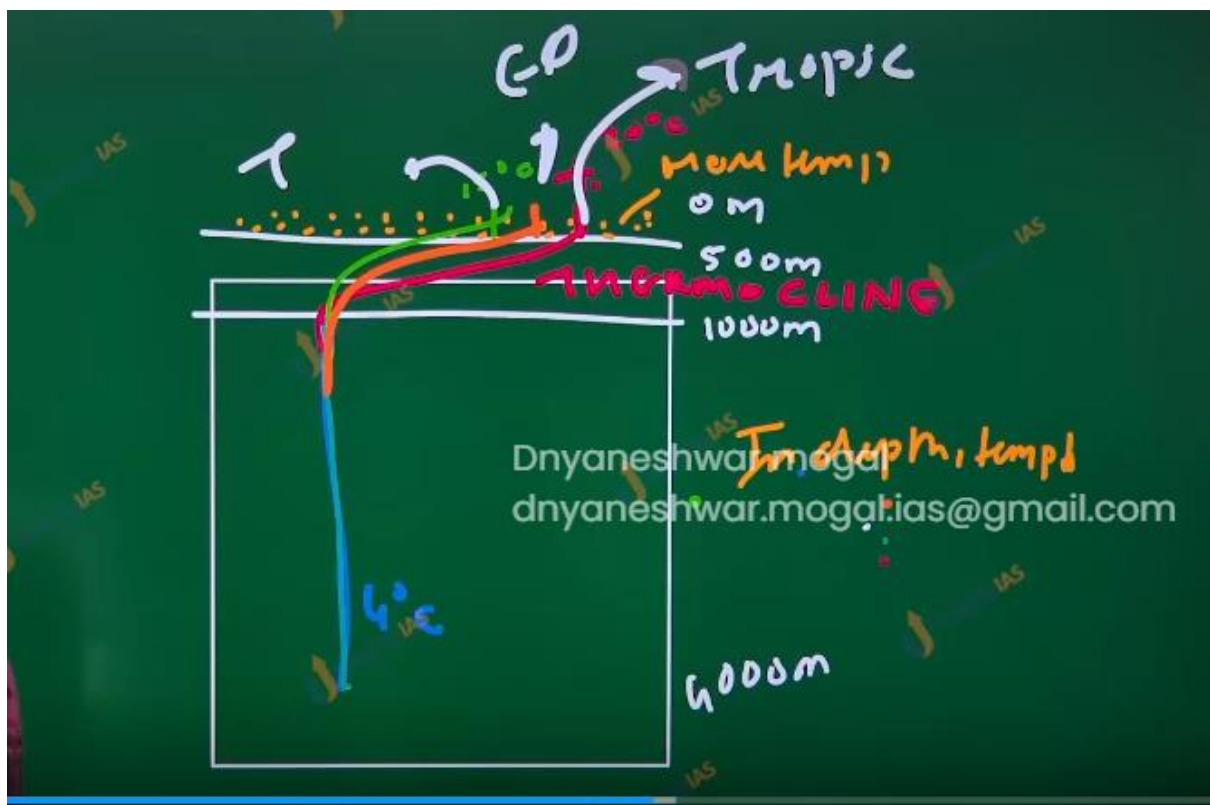
(v) Conditions in enclosed Sea

- The enclosed seas in the low latitudes record relatively higher temperature than the open seas;

Whereas

- The enclosed seas in the high latitudes have lower temperature than the open seas.

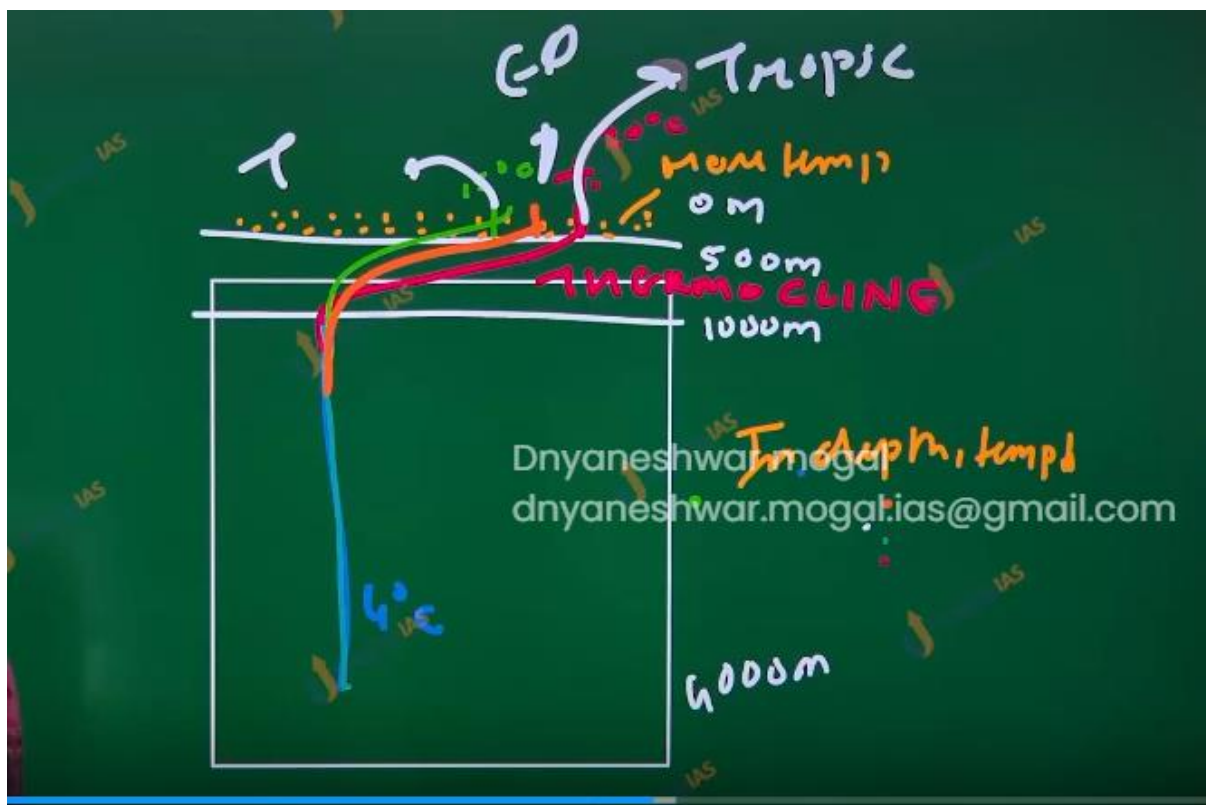




Pattern in the Ocean temperature.

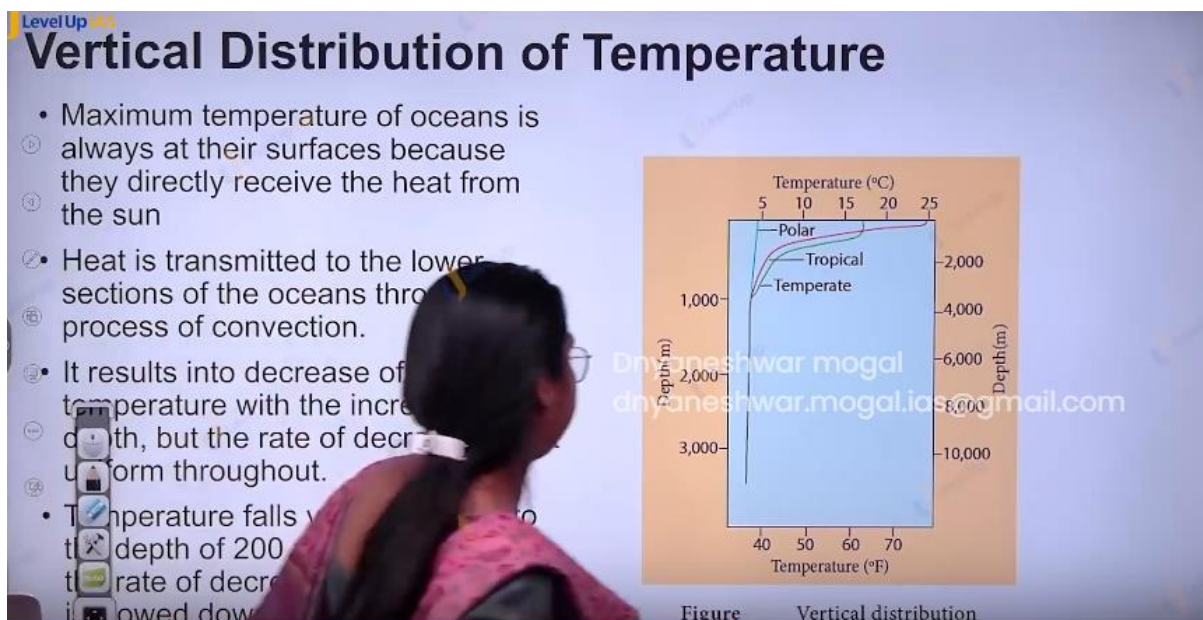
Vertical pattern

1. The vertical profile mentions how temperature decreases with increasing in depth.
 2. The maximum temperature of the ocean is on the surface. (generally, in the lower and mid latitudes) because they receive heat from the sun.
 3. Heat transmitted to the lower section of the ocean therefore temperature reduces with increasing in depth.
 4. But the rate of decrease is not the same throughout. The temperature drops very rapidly between 500m 1000m and thereafter rate of decrease of temperature is slowed down.
 - 5.
-



-
1. The first layer:
 - a. The top layer of warm oceanic water is 500 m thick.
 - b. The temperature is about 20-25 degrees Celsius.
 - c. This layer within the tropical region is present throughout but in mid latitude it developed only during the summer.
-

2. Second layer: (thermocline) extends from 500 to 1000 m.
 - a. There is rapid decrease in temperature with increased in the depth.
3. Third layer:
 - a. very cold and extends up to deep ocean floor
 - b. The temperature is nearly the same 4 degrees Celsius.
4. Arctic and Antarctic the surface water temperature can range from 0 - 4 degrees Celsius the temperature therefore the temperature change with change in depth is very less hence it can be said only one layer of cold water exist from the surface water to deep ocean floor.
- 5.



Vertical Distribution of Temperature

- Maximum temperature of oceans is always at their surfaces because they directly receive the heat from the sun
- Heat is transmitted to the lower sections of the oceans through the process of convection.
- It results into decrease of temperature with the increasing depth, but the rate of decrease is not uniform throughout.
- Temperature falls very rapidly up to depth of 200 m and thereafter, the rate of decrease of temperature is slowed down.

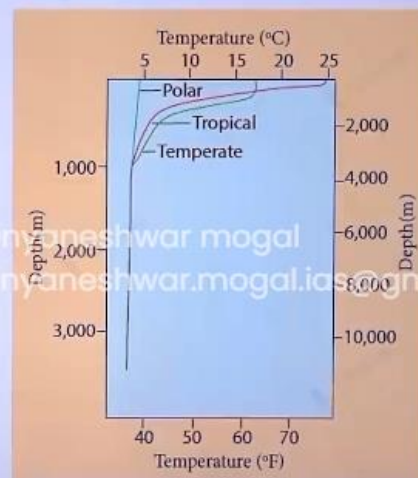


Figure Vertical distribution of temperature in Oceans

Vertical Distribution of Temperature

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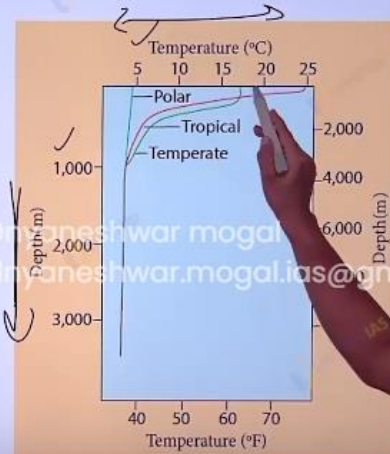


Figure Vertical distribution of temperature in Oceans

Horizontal distribution of temperature:

Horizontal Distribution of Temp:

1. **Average temperature** of surface water of oceans is **about 27°C**
2. It **gradually decreases from the Tropics towards the poles**.
3. Rate of decrease of temperature with increasing latitude is **about 0.5°C/ Latitude**
4. Average temperature is around 22°C at 20° latitudes, 14° C at 40° latitudes and 0° C near poles.
5. Means in **northern hemisphere (19° C) record higher temperature than in southern hemisphere (16° C)** due to unequal distribution of land and water
6. **Best temperature is not recorded at the equator but slightly towards north of it.**

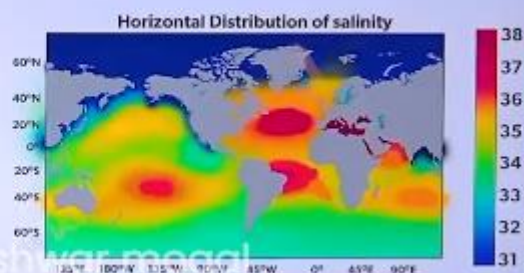


1. The average temperature of ocean is 25 to 27 degrees Celsius it decreases from the tropic to the poles. The rate of decrease with increase in latitude is .5 degree Celsius per latitude.
2. The temperature in the northern hemisphere is slightly more than southern hemisphere. Due to unequal distribution of land and water.
3. The maximum temperature is not recorded. At the equator but is recorded at the tropical location.
- 4.

Salinity of Ocean water.

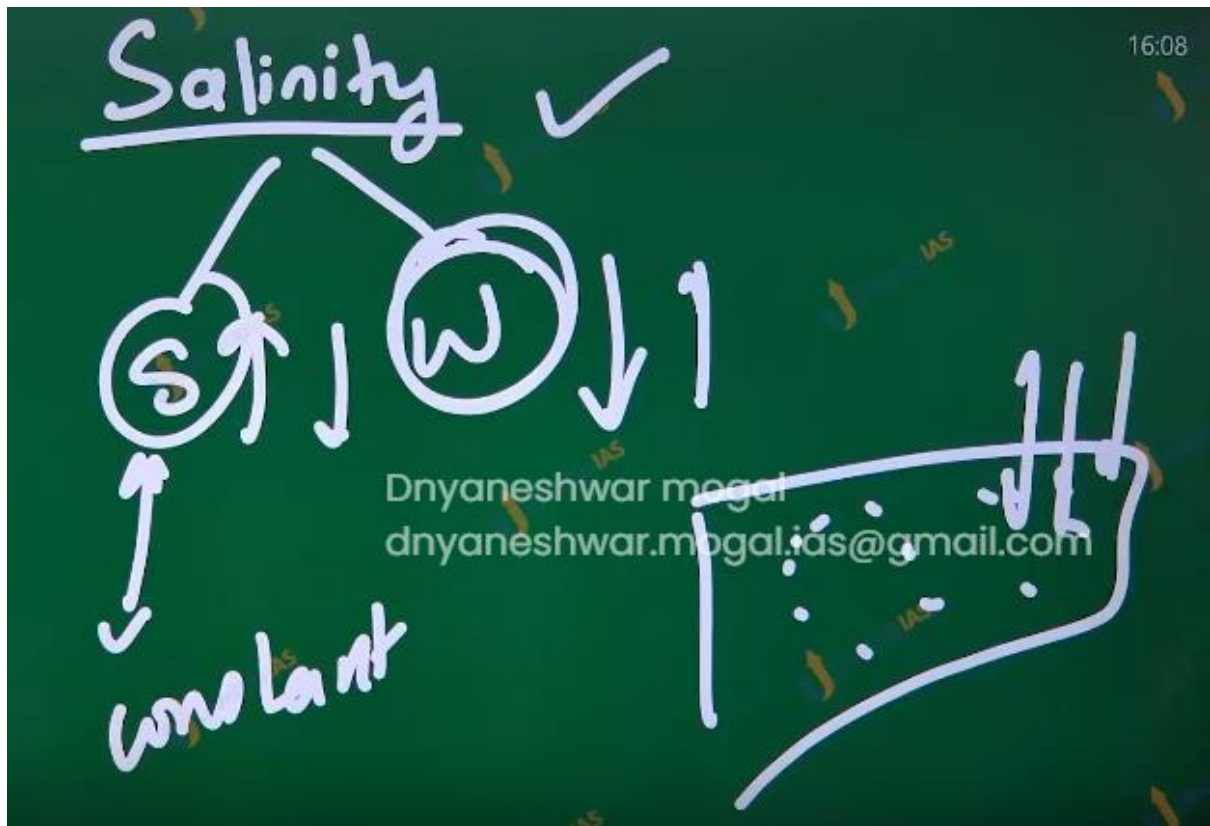
SALINITY OF OCEAN WATERS

- All waters in nature, whether rain water or ocean water, contain dissolved mineral salts.
- **Salinity:** Total content of dissolved salts in sea water.
- **Salinity:** Amount of salt (in gm) dissolved in 1,000 gm (1 kg) of seawater.
- It is expressed as parts per thousand or ppt.
- Salinity of 24.7 ppt demarcates upper limit for 'brackish water'.



Sources of Salinity

- Continents (rocks), through dissolving action of rivers and rainfall.
- Volcanic Eruptions on ocean floor eg: the Hotspots and MOR etc.
- Salts from rocks on ocean when geothermal active zone comes in contact with rocks
- Biological lifeforms





-
1. Salinity refers to the amount of salt (in grams dissolved in 1000 gm of sea water.) It is expressed as parts per 1000.
 2. The sources of salt in the ocean water are.
-
- a. Continental rock through dissolving the action of river and rainfall.
 - b. Volcanic eruptions of ocean floor.
 - c. Biological life forms
 - d. Salts from the rocks on the ocean floor when high geothermal winds dissolved them.
-



Factor affecting the salinity of ocean water.

1. Evaporation and precipitation:

- a. The salinity in the ocean floor decreases due to precipitation.
And increases due to evaporation.

2. Dilution factor:

- a. Salinity is influenced by freshwater flow from the rivers.
- b. Red sea vs Persian Gulf
- c. Persian Gulf has less salinity due to dilution by the water. By u-Fred and anti-grads
- d. The freezing and thawing:
 - i. Freezing and thawing of ice also affect the ocean salinity.
 - ii. Freezing increases salinity due to the process of brine rejection. (Ice is always formed by fresh water and therefore due to freezing of water the salinity of sub-surface water increases.

e. Thawing of water:

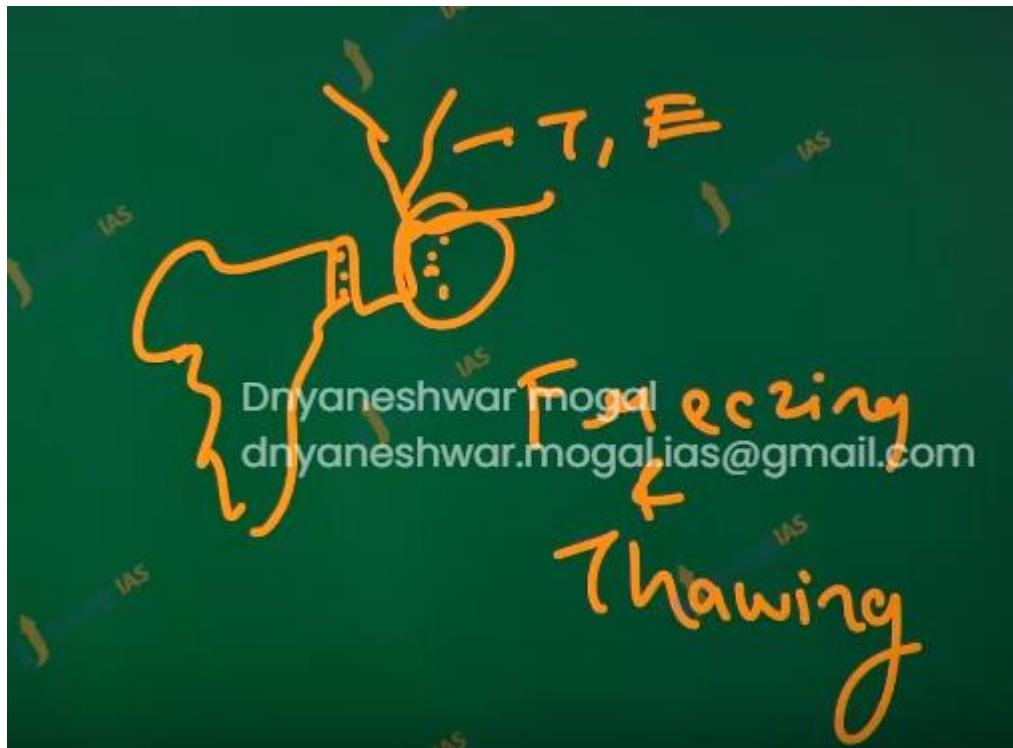
- i. Thawing reduces the salinity:
-

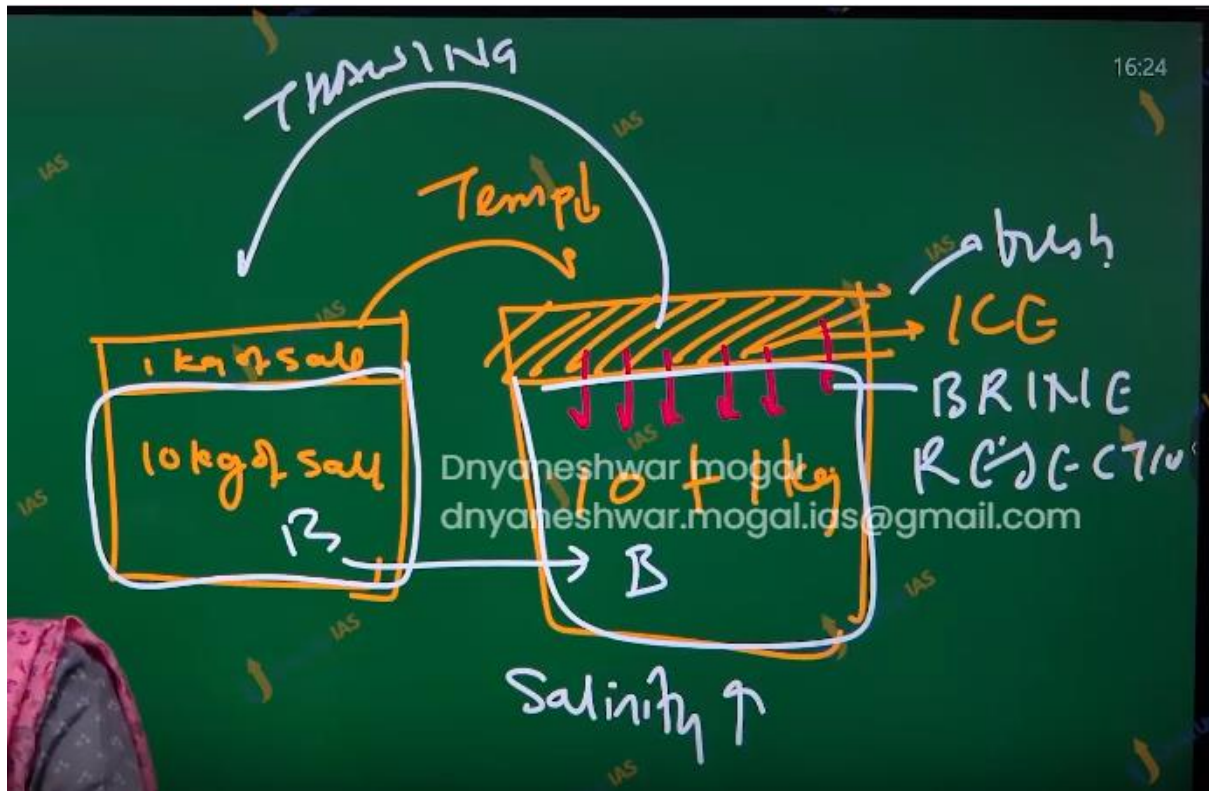
3. Winds:

- a. Wind influences the salinity of the water by affecting the evaporation.
- b. Winds also transfer the saline water from one place to another.

4. Ocean current:

- a. It contributes to salinity by influencing the temperature.
- b. Warm water dissolves the more salt than cold water therefore they have high salinity.





SALINITY OF OCEAN WATERS

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- Salinity: Total content of dissolved salts in sea water.
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- Biological lifeforms

Horizontal Distribution of salinity

Factors affecting salinity of ocean waters

1. **Evaporation and precipitation:** Salinity of water in the surface layer of oceans depend mainly on evaporation and precipitation.
 2. **Dilution Factors:** Surface salinity is influenced by fresh water flow from rivers along , and in polar regions by the processes of freezing and thawing of ice.
- Wind:** Influences salinity of an area by transferring water to other areas and also affects evaporation
- Ocean currents** contribute to the salinity variations. Warm Water dissolves more salts in general.

Highest salinity in water bodies
Lake Van in Turkey (330‰),
Dead Sea (238‰),
Great Salt Lake (220‰)

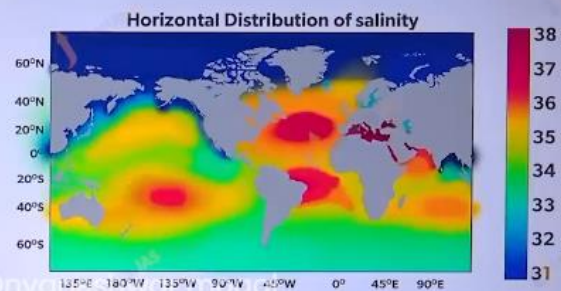
Dnyaneshwar mogal
dnyaneshwar.mogal.ias@gmail.com

HORIZONTAL DISTRIBUTION OF SALINITY

- Salinity for normal open ocean: 33 ppt- 37 ppt
- Land locked Red Sea: 41 ppt
- Estuaries and Arctic, salinity: 0-35 ppt. Hot and dry regions, evaporation is high, salinity sometimes reaches to 70 ppt

Salinity in Pacific Ocean

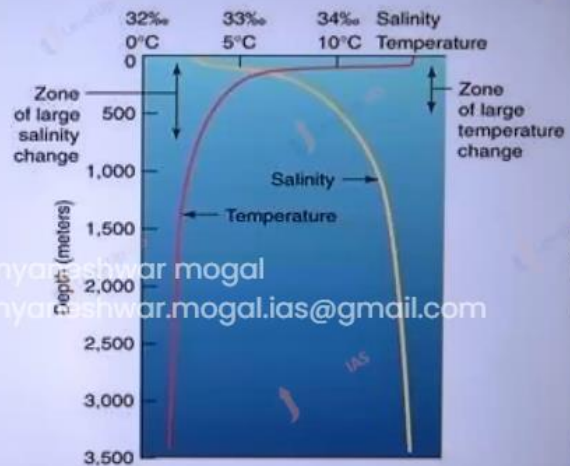
- Salinity variation in Pacific Ocean is mainly due to its shape and larger areal extent.
- Salinity decreases from 35 ppt (around the equator) - 31 ppt on the western parts of the northern hemisphere because of the influx of fresh water from the Arctic region.
- In the southern hemisphere, it decreases to 33 ppt.



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Vertical Distribution of Salinity

- Salinity changes with depth
 - Salinity at the surface
 - Increases by loss of water to ice or evaporation, or
 - Decreased by the input of fresh waters, such as from the rivers.
 - Salinity at depth is very much fixed, as there is no way that water is 'lost', salt is 'added.'
- There is marked difference in salinity between surface and deep zones of oceans.



Lets go beyond NCERT

Additional Information Here

1. Salinity Pattern
2. Coral Reef



Factors affecting salinity of ocean waters

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 Dead Sea (238 ‰),
 Great Salt Lake (220 ‰)

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Pattern of salinity

Horizontal and vertical.

Hemispheric contrast:

1. Northern waters are more saline as compared to southern water.
2. The Equator does not have the highest salinity. Highest salinity is recorded along the tropics.
3. In general salinity decreases from tropics to pole because water becomes cooler towards the poles.
4. Warm enclosed and semi enclosed water of tropical area will have higher salinity than the open ocean. Ex red sea, Mediterranean Sea.
5. However the tropical water receiving the freshwater from the river will have less salinity, ex. Persian Gulf vs red sea.
6. A small lake with high evaporation has extremely high salinity.
7. Ex lake warn in turnkey 330 part per 1000, dead sea 238 part per thousand.
8. Ocean waters that receive warm current will have higher salinity. Ex. Norwegian sea.

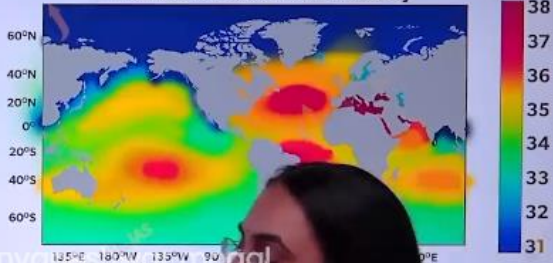
Cold water with upwelling has lower salinity. Ex. Norwegian Sea

HORIZONTAL DISTRIBUTION OF SALINITY

- Salinity for normal open ocean: 33 ppt- 37 ppt
- Land locked Red Sea: 41 ppt
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Salinity in Pacific Ocean

- Salinity variation in Pacific Ocean is mainly due to its shape and larger areal extent.
- Salinity decreases from 35 ppt (around Tropics) - 31 ppt on the western parts of the northern hemisphere because of the influx of melted water from the Arctic region.
- After 15° - 20° south, it decreases to 33 ppt.



Horizontal Distribution of salinity

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Salinity in Atlantic Ocean

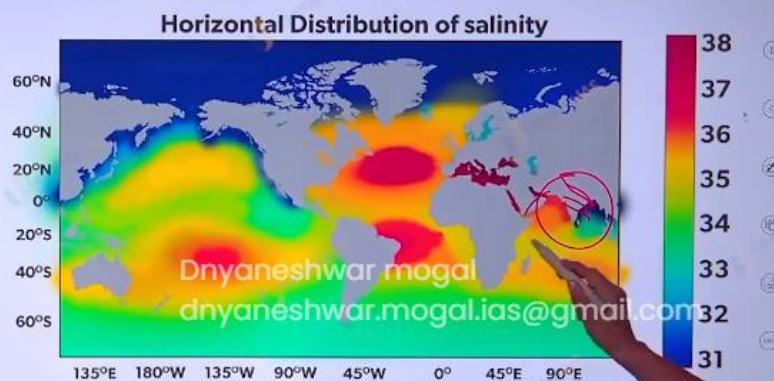
- Average salinity of Atlantic Ocean: 36 ppt.
- Highest salinity: 15° & 20° latitudes around 37 ppt.
- It gradually decreases towards the north.
- North Sea, in spite of being in higher latitudes, records higher salinity due to more saline water brought by the North Atlantic Drift.
- Baltic Sea records low salinity due to influx of river waters in large quantity.
- The Mediterranean Sea records higher salinity due to high evaporation.
- Salinity is, however, very low in Black Sea due to enormous fresh water influx by rivers.



Figure 13.5 : Surface salinity of the World's Oceans

Salinity in Indian Ocean

- The average salinity of the Indian Ocean is 35 ppt
- The low salinity trend is observed in the Bay of Bengal due to influx of river water. On the contrary, the Arabian Sea shows higher salinity due to high evaporation and low influx of fresh water

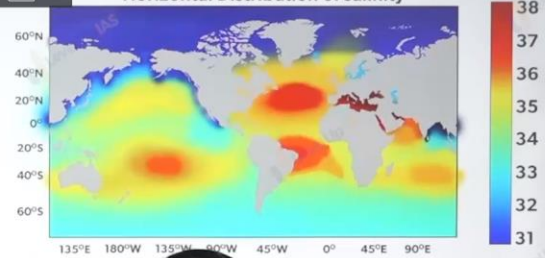


Ocean current video

SALINITY OF OCEAN WATERS

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Horizontal Distribution of salinity



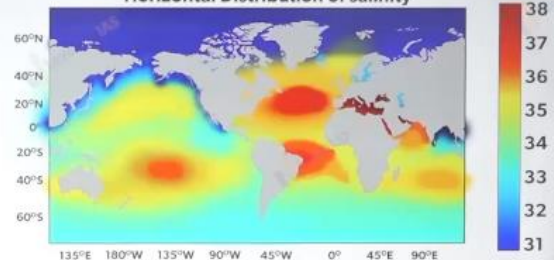
- All waters in nature, whether rain water or ocean water, contain dissolved mineral salts.
- Salinity: Total content of dissolved salts in sea water.
- Salinity: Amount of salt (in gm) dissolved in 1,000 gm (1 kg) of seawater.
- It is expressed as parts per thousand or ppt.
- Salinity of 24.7 ppt demarcates upper limit for 'brackish water'.

Sources of Salinity

- Continents (rocks) through dissolving action of rivers and rainfall.
- Volcanic Eruptions on ocean floor eg: the Hotspots.
- Salts from rocks on ocean when geothermal active zone comes in contact with rocks.
- Biological lifeforms.

SALINITY OF OCEAN WATERS

Horizontal Distribution of salinity



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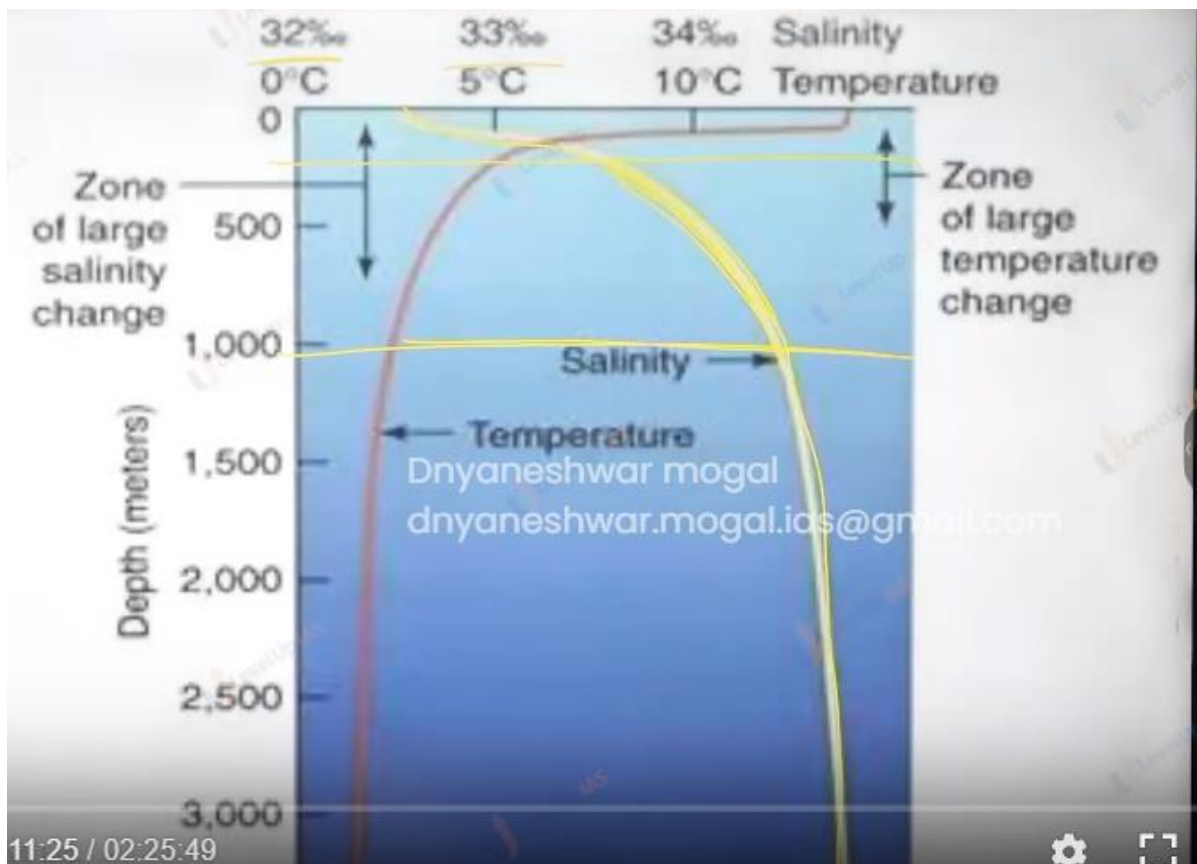
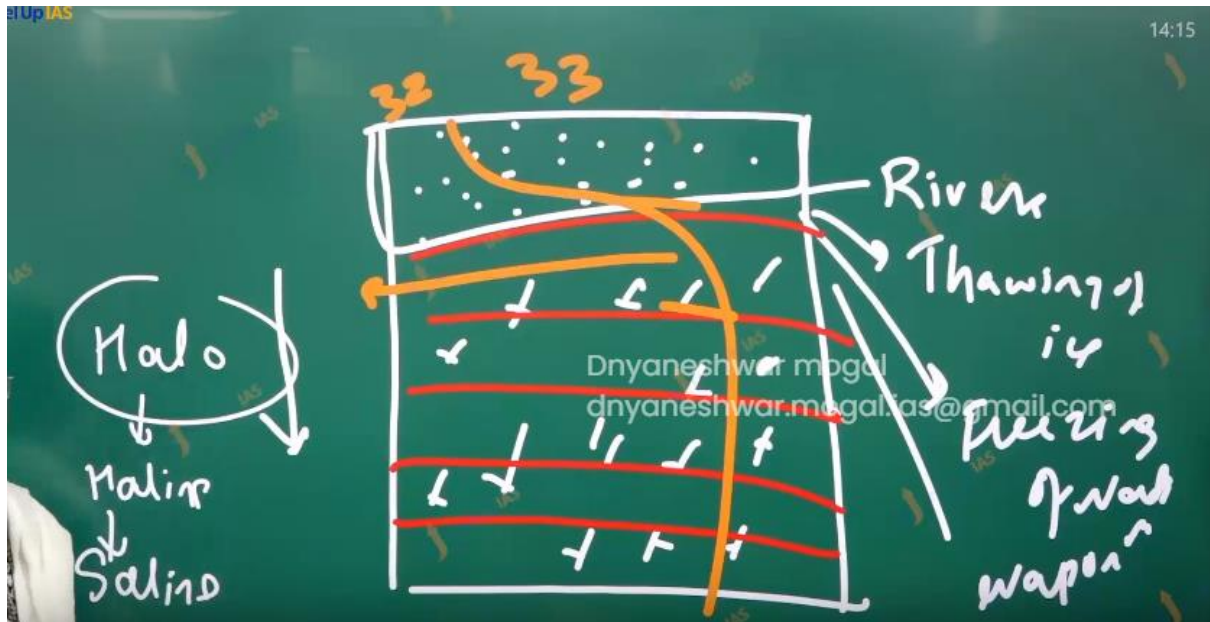
Factors affecting salinity of ocean waters

- 1. Evaporation and precipitation:** Salinity of water in the surface layer of oceans depend mainly on evaporation and precipitation.
- 2. Dilution Factors:** Surface salinity is influenced by fresh water flow from rivers along , and in polar regions by the processes of freezing and thawing of ice.
- 3. Wind:** Influences salinity of an area by transferring water to other areas and also affects evaporation
- 4. Ocean currents** contribute to the salinity variations. Warm Water dissolves more salts in general.

Highest salinity in water bodies
Lake Van in Turkey ($330^{\circ}/_{\infty}$),
Dead Sea ($238^{\circ}/_{\infty}$),
Great Salt Lake ($220^{\circ}/_{\infty}$)

Vertical pattern in the ocean salinity.

1. Salinity changes with the depth. The surface water salinity can change due to many factors. Such as melting of ice delusion of river, loss of water evaporation, freezing of water.
2. Salinity of depth is more or less fixed there is no way drastic change in salt added or removed.
3. Lower saline water rests over higher saline water. Therefore, salinity generally increases with depth.
4. The sharp increase in the salinity is 500m 1000m this zone is called as halocline.
5. The high saline sea water generally sinks below the lower saline waters. This leads to stratification by salinity.



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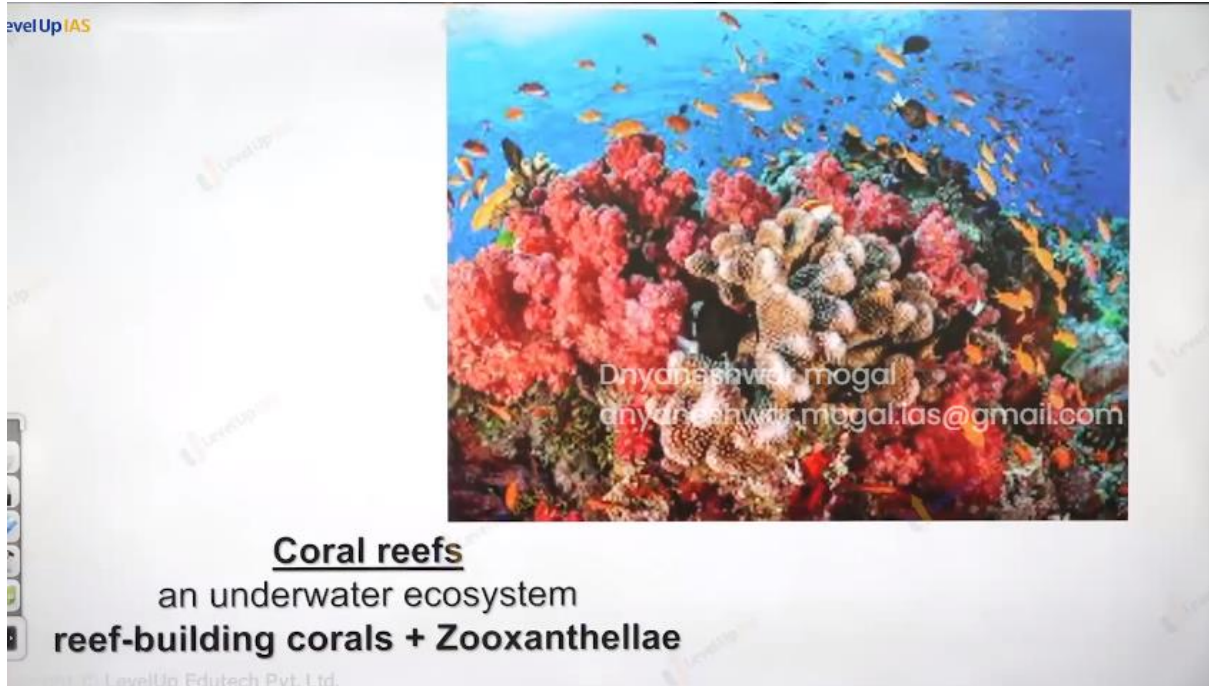
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(25) Ocean Currents

Coral polyp: (it is an animals)

Coral light: it is skeletal remain of the coral column.

Coral reefs: accumulation of skeletal remain result in the coral reefs in simple term corallite accumulation to formed coral reefs.

-
1. Coral polyps: lives in symbiotic relationships with algae called zooxanthellae.
 2. This symbiotic relationship is very important ecologically as coral polyp provides shelter to zooxanthellae. And zooxanthellae provide the food the coral polyps.

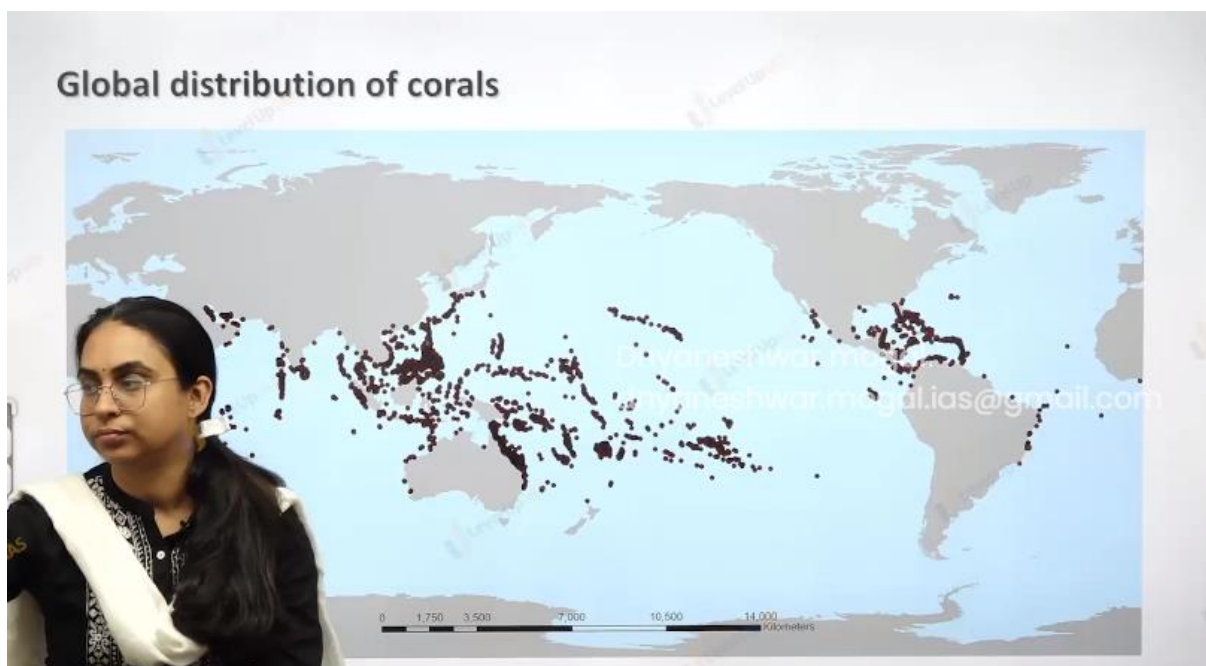
3. When the coral polyps die, the skeletal remain cumulate as coral reefs.
 4. Coral reefs are very important ecosystems. They are the rainforest of the ocean. High carbon sequestration (meaning ability to observe) potential.
 5. They facilitate rich biodiversity, they are rich fishing grounds and facilitate marine tourisms.
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Important condition conducive for coral reefs and coral polyps

1. Sea surface temperatures 23 28 degree Celsius. Therefore, they grow in the tropical ocean.

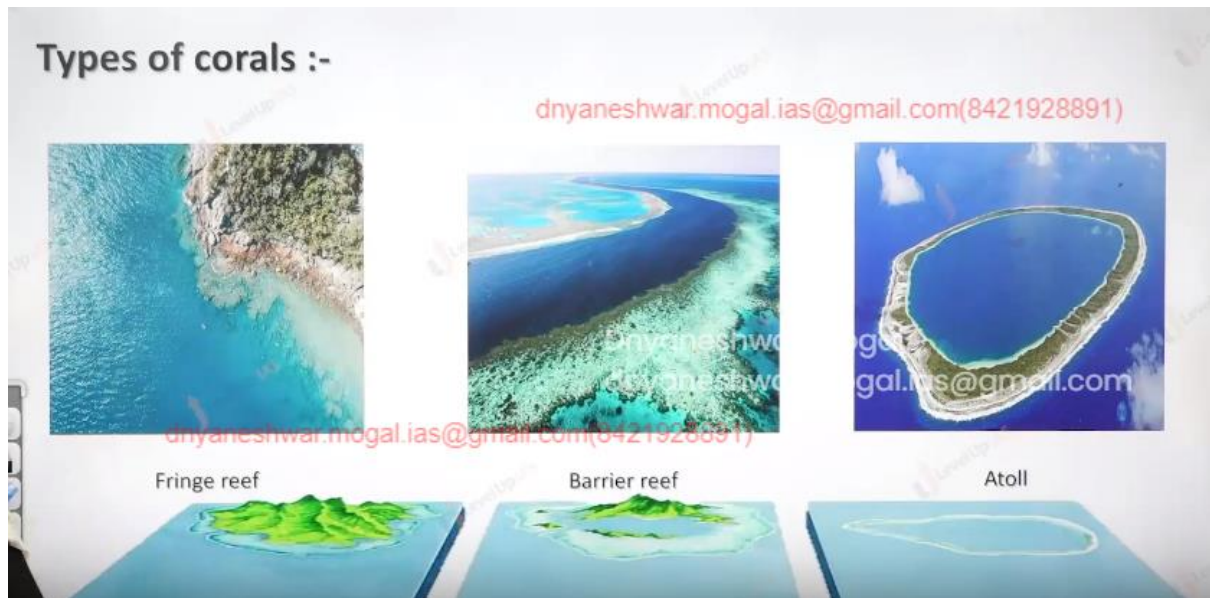
2. Salinity to be around the 30 35 parts per 1000. They can survive near the river mouth.
 3. Clean water (absent of muddy water as they can block the pores of coral reefs they need low nutrients water. They can't survive the high enriched water.
 4. They need a platform which is shallow. Coral polyps cannot swim can't float and therefore they need the platform for survival.
 5. Platform should be shallow within the photic zone as zooxanthella needs sunlight for photosynthesis.
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Distribution of coral reefs:

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1. They are found along the Gulf of Mexico, Caribbean Sea, western Atlantic Ocean, Red Sea, Persian Gulf, Andaman Nicobar Island. There

is coral triangle along northeast Australia, Indonesia, Philippines majority of pacific island has coral reefs.



Type of corals reefs:

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1. Coral reefs are landform therefore they are remains of coral skeletal.
 2. The 3 main types of coral reefs are fringe reef, barrier reef , atoll.
 - 3.
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Types of corals :-



Fringe reef



Barrier reef



Atoll



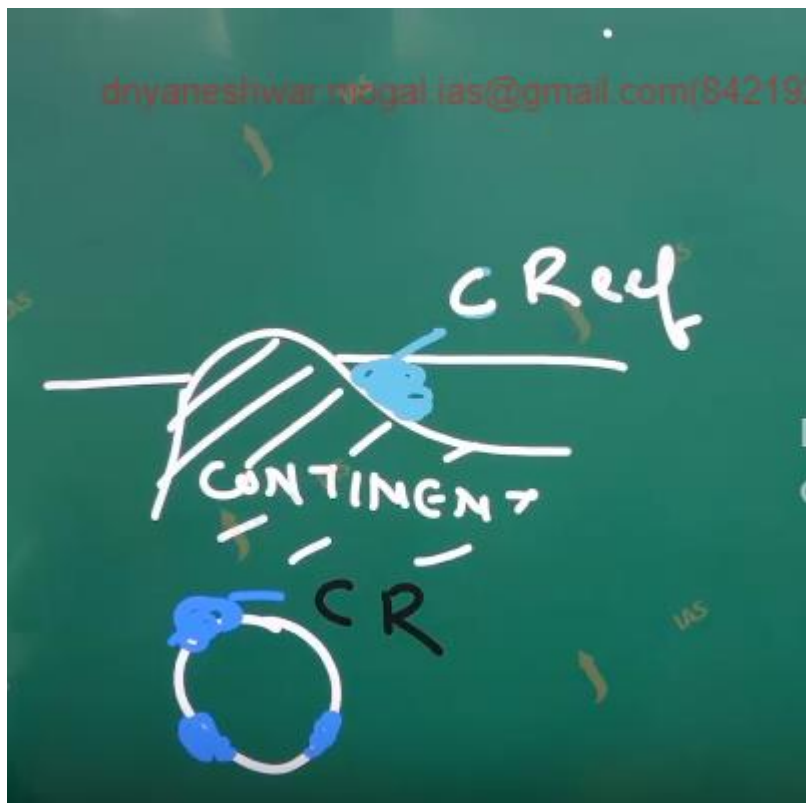
Fringing Reef



Barrier Reef



Atoll



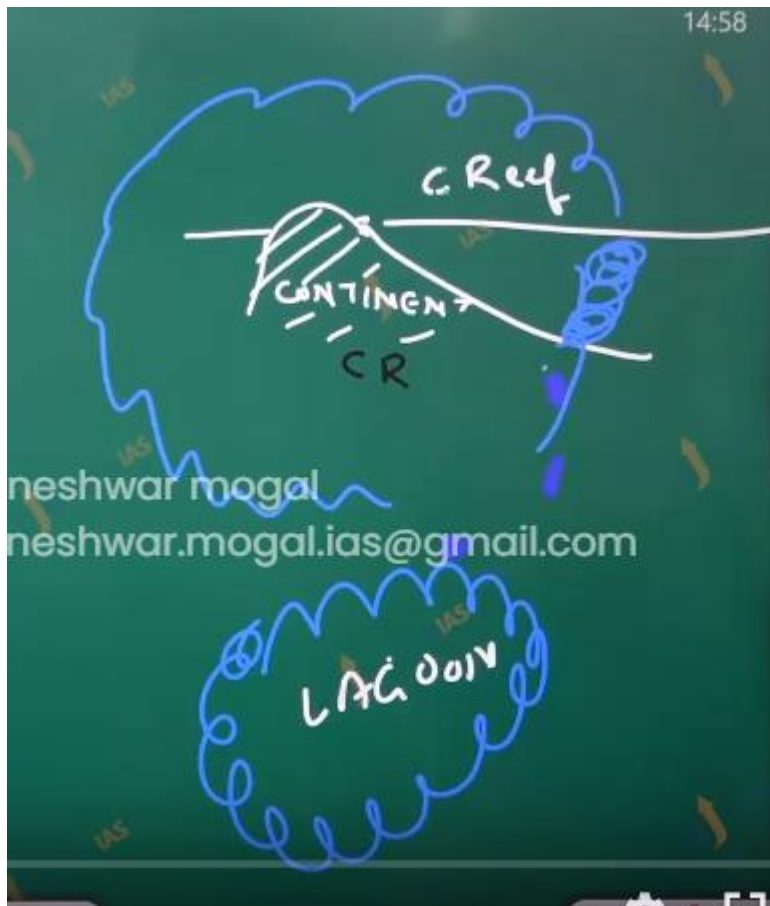
Fringe reefs:

1. They are on the margin of the continent.
-



Barrier reefs:

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1. They are wide elongated extensive coral reefs, and they are further away from the land
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Atoll:

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1. They are circular, coral reefs formations, they are very farer from the land and they can enclose shallow lagoon(meaning lake).
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Consider the following statements

1. Most of the world's coral reefs are in tropical waters.
2. More than one-third of the world's coral reefs are located in the territories of Australia, Indonesia and the Philippines.
3. Coral reefs host far more animal phyla than those hosted by tropical rainforests.

Which of the statements given above is/are correct?

- (a) 1 and 2 only
- (b) 3 only
- (c) 1 and 3 only
- (d) 1, 2 and 3

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Answer is d.

Most coral reefs are found in the tropical water.

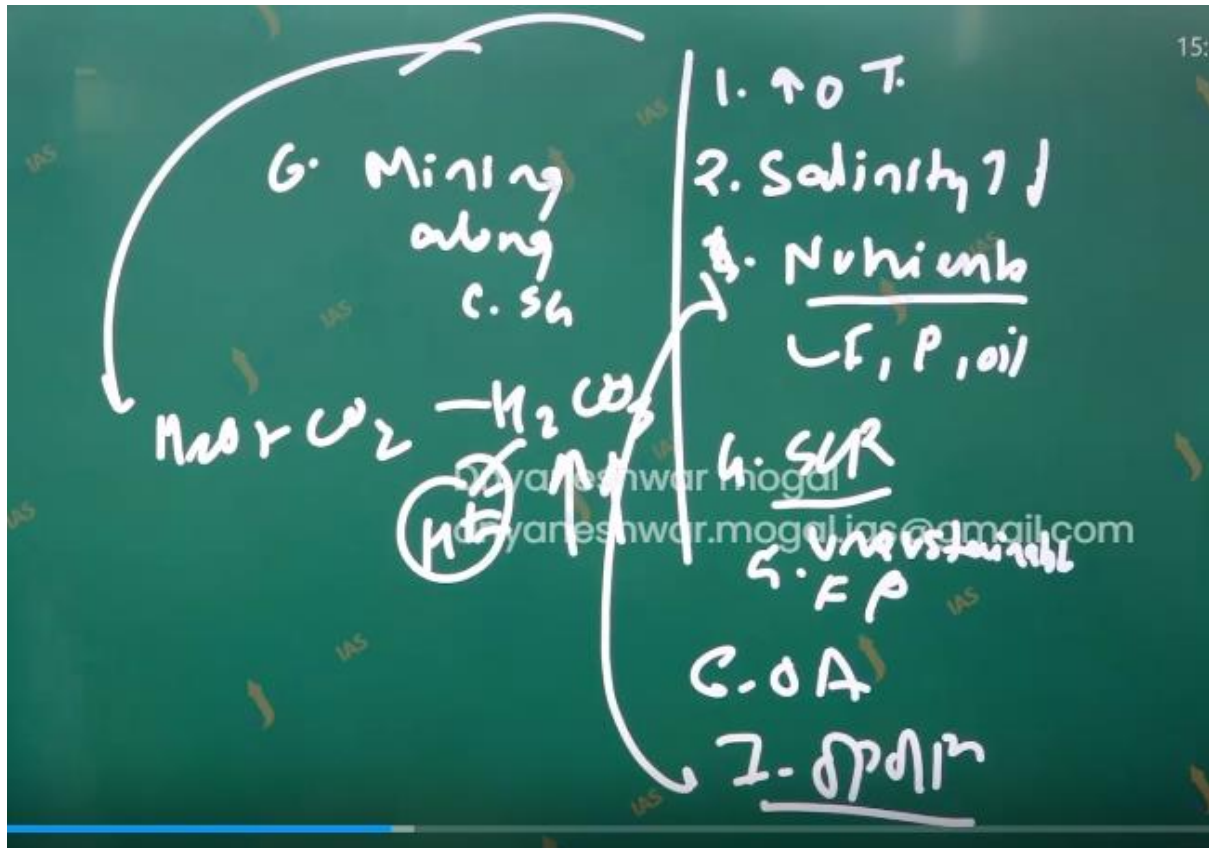
1/3 rd. coral reefs are found in Indonesia, Philippines and Australia.

Coral reefs store far more animal phyla than the tropical rainforest.

Bio rock technology:

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1. It aims to build the artificial reefs by the applying low voltage electrical current in the ocean to facilitate the dissolved minerals to accumulate on the some platform.
 2. The aim of making artificial reefs is to create the habitat for new coral polyps to make symbiotic relationships with zooxanthellae.

3. The technology is used by Zoological survey of India with help of Gujarat Forest department
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Coral bleaching:

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1. It refers to the loss of symbiotic relation between coral polyps and zooxanthellae.
 2. The coral reefs are very colorful structure but due loss of symbiotic relation the coral loss their color causing the bleaching.
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Reason behind the beaching.

-
1. Global warming / increasing the ocean temperature: symbiotic relation can survive only in the narrow range of temperatures.
 2. Ocean acidification: increase in the H positive ions can cause dissolution of coral shell of coral polyps.
 3. Changes in the salinity:
 4. Increase in the sedimentation due to mining activity.
 5. Water pollution due to fertilizer and pesticide, oil spills.
 6. Unsustainable fishing practices bottom trawling, cyanide fishing.
 7. Increase in the diseases like black band diseases virus attack, virus Shiloh attacks in the Mediterranean Sea.
 8. Event like El Nino El Nina
 9. Increase storms and cyclones
 10. Ex.
 - a. Between 2014-2017 El Nino the Great Barrier Reefs was bleached by 50 %.
 - b. Between 1997-1998 90% of coral reefs of North Indian Ocean were destroyed.
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Mains Question

Q) Assess the impact of **global warming** on the **coral life system** with examples. **2019 UPSC Question**

Q) Account for variations in oceanic salinity and discuss its multi-dimensional effects. **2017 UPSC Question**

for variations in oceanic salinity and discuss its multi-effects. **2017 UPSC Question**

-
1. GW $\rightarrow$ POT $\rightarrow$ Bleach
 2. GW $\rightarrow$ Ocean chem. res. $\rightarrow$ loss of ch.
 3. GW $\rightarrow$ sublimation, + wind to h-h evap $\rightarrow$ + salt $\rightarrow$

(25) Ocean Currents
